

JAMNAGAR MUNICIPAL CORPORATION

TENDER FOR

Design, Development, Implementation, and Maintenance of
Spatially enabled Web Portal using latest Technology for
Jamnagar Municipal Corporation

TENDER NOTICE (ONLINE) NO.

VOLUME: I

**Invitation for Bid, Instructions to Bidders,
General Conditions of Contract, Contract Data, &
Qualification Information, and other forms**

Downloading of Tender Documents	From Date : 24 - 03 - 2025 16:00 Hrs. to Date :15-04-2025 18:00 Hrs.
Pre-bid Meeting Venue : Date and Time	From Date: 04-04-2025 12:00 Hrs. Deputy Engineer's Office, Project & Planning Department 2nd Floor, New Admin. Bldg., Lal Bungalow, Jamnagar
TECHNICAL BID OPEN	From Date: 25-04-2025 12:00 Hrs.
Submission of EMD, Tender Fee & Other documents as per qualification criteria and technical bid form.	From Date : 16-04-2025 to 22-04-2025 during Office Hours in the Office of The Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001 by Speed Post / RPAD in sealed cover duly super scribed with name of work. [By Courier OR Personally Documents will not be accepted]
PRICE BID OPEN	From Date: 02-05-2025 12:00 Hrs.

The Municipal Commissioner,
Jamnagar Municipal Corporation,
Jubilee Garden,
Lal Bungalow Circle,
Jamnagar, Gujarat.
Phone : 0288-2550231 Ext. 541
Fax : 0288-2554454
Email : jmc.project27@gmail.com

Design, Development, Implementation,
and Maintenance of Spatially enabled Web
Portal using latest Technology for
Jamnagar Municipal Corporation

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1 Invitation of Bid

Jamnagar Municipal Corporation invites “Development, Implementation, and Maintenance of Existing spatially enabled Web Portal using latest Technology for Jamnagar Municipal Corporation” as per the terms and conditions attached with this document.

Note: Bidder/Agencies are advised to study this RFP document carefully before submitting their proposals in response to the RFP Notice. Submission of a proposal in response to this notice shall be deemed to have been done after careful study and examination of this document with full understanding of its terms, conditions, and implications.

1.1 Project Background

Identification of urban landscape in developed cities provide unsurpassed assistance in improving critical city focused activities such as emergency and disaster management, citizen engagement, planning, zoning and development. The potential to assist planners in reaching local climate resilience, economic development and housing goals.

Leveraging these international best practices, Jamnagar Municipal Corporation intends to perform a holistic asset inventory with spatial information for above and underground assets, develop, implement & maintain Web and Mobile applications, with change detection module on city, year-on-year basis.

1.2 Project Objective

JMC has set following objectives of the Project:

- 1) Collection of Ground Control Points (GCPs)
- 2) Imagery Acquisition 30-50 CM from 1990-2024
- 3) Preparation of 2D layers as per AMRUT Guidelines
- 4) Scanning of underground utilities
- 5) Assets Inventory of Jamnagar capturing all changes every year for 3 years after Go-Live
- 6) Development Implementation of Spatially enabled Web Application.

- a. Department and Citizen Web Application (2D and 3D compatible)
 - b. Assets Inventory Monitoring Web application with analytics for Change Detection monitoring
 - c. Spatially enabled Web Dashboards
 - d. Integration of web application & services with existing IT & other applications
- 7) Development of Property Assessment solution including Mobile and Web Application
 - 8) Security testing, by CERT-IN empaneled agency at Go-Live & then every year for 3 years
 - 9) Hosting of data on JMC storage.
 - 10) Setting up of GIS Cell
 - 11) Annual Maintenance Contract of all data, applications & services provided under the scope of work for 3 years after Go-Live.

The details of each objective are explained in the scope of work section.

1.3 Bid Document Notice

- 1) This bid Document is being published by The Commissioner, Jamnagar Municipal Corporation, for the selection of System Integrator (SI) for Development, Implementation, and Maintenance of Existing spatially enabled Web Portal using latest Technology for Jamnagar Municipal Corporation.
- 2) Bidder agency is advised to study this bid document carefully before submitting their bids in response to the Bid Notice. Submission of a bid in response to this notice shall be deemed to have been done after careful study and examination of this document with full understanding of its terms, conditions, and implications.
- 3) This bid document is not transferable.
- 4) The complete bidding document has been published on 24-03-2025 at 16:00 View for the purpose of downloading. The downloaded bidding document shall be considered valid for participation in the electronic bidding process (e-

Tendering) subject to the submission of required tender/ bid document fee and EMD.

- 5) Bidders who wish to participate in this bidding process must register on 24-03-2025.
- 6) To participate in online bidding process, Bidders must procure a Digital Signature Certificate (Class - II) as per Information Technology Act-2000 using which they can digitally sign and encrypt their electronic bids. Bidders can procure the same from any CCA approved certifying agency, i.e. TCS, Safecrypt, Ncode, etc. Bidders who already have a valid Digital Signature Certificate (DSC) need not procure a new DSC. Please refer Bidders manual on JMC portal for further details.
- 7) A Three-envelope selection procedure shall be adopted.
- 8) Bidder (authorized signatory) shall submit their offer online in electronic formats of technical (including prequalification documents) and financial proposal.
- 9) JMC will not be responsible for delay in online submission due to any reason. For this, bidders are requested to upload the complete bid proposal, well advance in time so as to avoid issues like slow speed, choking of web site due to heavy load or any other unforeseen problems. For queries related to e-Tender, kindly contact <https://tender.nprocure.com/>.
- 10) Bidders are also advised to refer “Bidders Manual” available at <https://tender.nprocure.com/> for further details about the e-tendering process.

Director, Information Technology, Jamnagar Municipal Corporation invites bidders to submit their technical bids and financial offers for the selection of System Integrator (SI) for Design, Development, Implementation, and Maintenance of Spatially enabled Web Portal using latest Technology for Jamnagar Municipal Corporation in accordance with conditions and manner prescribed in this bid document.

2 Pre-Qualification Criteria

Any entity that meets the below defined Pre-qualification criteria can participate in the bid process. Entities desirous of participating in the bid process need to pass through the three-stage tender process each with its own qualification criteria.

#	Qualification Criteria	Documentary Evidence
1	The Sole bidder must be a company registered under Indian Companies Act, 2013 since last 10 Years	Copy of Certificate of Incorporation signed by Authorized Signatory of the Bidder
2	The Sole Bidder should have an average annual turnover of at least Rs 25 Crore from Remote Sensing/ Photogrammetry/ GIS services/ over Three financial years.	Copy of the audited Balance Sheet and Profit & Loss Statement of the company and/or Certificate from the Chartered Accountant clearly stating the average annual turnover Remote Sensing/ Photogrammetry/ GIS services/ over Three financial years.
3	The Sole Bidder should individually have positive net worth in the last three financial years. Also, the Sole bidder should not have individually declared loss in last five financial years	Audited balance sheet, profit & loss statement and certificate from certified Chartered Account mentioning positive net worth.
4	The Sole Bidder must have executed projects for Smart cities /Municipal Corporations/ Government Agencies containing mapping for the cities of below mentioned value in last five financial years: 1. Two projects with a value of not less than ₹ 2 Crore each	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope
5	The Sole Bidder should have developed COTS COTSbased Web GIS platform for Smart cities/Municipal Corporations/Government Agencies) projects in last five years with a minimum value of Rs 2Cr	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope

6	The Sole Bidder should have experience of Implementation and Maintenance of Change detection of the Urban landscape Smart cities/Municipal Corporations/Government Agencies) in last three years with a minimum value of Rs 1.5 Cr	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope
7	The Sole bidder possess the below certification: • CMMi level 3 or higher • ISO 9001:2015 • ISO 27001:2022	Copy of valid certificate/s to be provided
8	Bidder should have integrated GIS with ICCC software with ULB/ Smart City / State government department in India	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope
9	The Sole bidder must have GST registration number.	Copy of GST registration number.
10	The Solebidder must not have been blacklisted or debarred by any Central/ State Government Organization in India at the time of submission and at the stage of signing the contract.	Declaration by the Bidder as per format given in the bid document
11	The Sole bidder must have at least 25full time technical resources in its payroll in the domain of software development/ GIS /Survey with qualification Diploma/B.E /B.Tech/M.E /M.Tech, BSc/MSc, BCA /MCA as on date.	Certificate from HR Head on the company letter head to this effect

3 Instruction to Bidder

3.1 Bid Data Sheet

#	Information	Details
A. Introduction		
1.	Project Name	Design, Development, Implementation, Maintenance of Change Detection using Geospatial Technology for JMC
2.	Bid Document reference No and Start Date	24-03-2025 at 16:00
3.	Bid Type	Open Tender via e-Tendering system
4.	Bid Document Price (Non- refundable)	18,000/-
5.	Bid Download End Date	15-04-2025 at 18:00
6.	Earnest Money Deposit (EMD)	8,50,000/-
7.	EMD Submission Due Date & Time	From Date : 16-04-2025 to 22-04-2025 during Office Hours in the Office
8.	Addressee & Address for EMD to be submitted	Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001 by Speed Post / RPAD in sealed cover duly super scribed with name of work. [By Courier OR Personally Documents will not be accepted]
9.	Bank Solvency Certificate	3,00,00,000/-
10.	Performance Bank Guarantee	3% of Total Contract Value(Exclusive GST)
B. Preparation of Bids		
11.	Language of Bid	English

12.	Bid Validity Period	180 Days
13.	Performance Security Validity Period	3 Months After Project Completion
C. Bid Presentation		
14.	Last date for submission of written queries for clarifications	04-04-2025
15.	Query Submission	04-04-2025
16.	Place, Date and Time of pre-bid meeting (Online/ Onsite)	Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001
17.	Contact Person for clarification of Queries	Deputy Engineer, PROJECT & PLANNING BRANCH
18.	Contact Information	jmc.project27@gmail.com
19.	Last date (deadline) for receipt of proposals in response to Bid Document notice	15-04-2025
20.	Place, Time and Date of opening of technical proposals (Packet A & B) received in response to the Bid Document notice	25-04-2025 AT 12:00 Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001
21.	Place, Time and Date of technical presentations by qualified bidders.	25-04-2025 AT 12:00 Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001

22.	Place, Time and Date of opening of Financial proposals received in response to the Bid Document notice	02-05-2025 AT 12:00 Deputy Engineer, PROJECT & PLANNING BRANCH, 2nd Floor, MahanagarSeva Sadan-1, Jamnagar – 361001
D. Evaluation of Bids and Awarding of Contract		
23.	Signing of Contract Agreement with JMC	-

3.2 Purpose of Bid Document

The purpose of this Bid Document is to select Agency for Development, Implementation and Maintenance of Change Detection using Geospatial Technology for JMC as per the scope defined in the subsequent sections. This document provides information to enable the bidders to understand the broad requirements to submit their 'Bids'.

3.3 Bidder Registration and Instructions

1. The complete bidding document will be published on 24-03-2025 for the purpose of downloading. The downloaded bidding document shall be considered valid for participation in the electronic bidding process (e-Tendering) subject to the submission of required tender/ bidding document fee and EMD.
2. Bidders who wish to participate in this bidding process must register on 24-03-2025, if not registered already. To participate in online bidding process, Bidders must procure a Digital Signature Certificate (Class – III) as per Information Technology Act-2000/2008 using which they can digitally sign and encrypt their electronic bids. Bidders can procure the same from any CCA approved certifying agency, i.e. Safescrypt, Ncode, etc. List of same is available on <https://cca.gov.in>.
3. A Three cover selection procedure shall be adopted.
4. The Fee/Pre-qualification related documents shall be submitted in Cover 1/Packet-A, the Technical Documents in Cover 2/Packet-B and the financial bid in Cover 3/Packet C or Item Data.

5. Bidder (authorized signatory) shall submit their offer online in electronic formats for preliminary qualification, technical and financial proposal. JMC shall not be responsible for delay in online submission by bidder due to any reason. For this, bidders are requested to upload the complete bid proposal well in advance so as to avoid issues like slow speed, choking of web site due to heavy load or any other unforeseen problems.
6. Bidders are also advised to refer “Bidders Manual” available at 24-03-2025 for further details about the e-tendering process.
7. Tender should be submitted online on or before due date as mentioned in RFP. The Financial Bid shall be opened of Technically Qualified bidders & they may attend the Tender opening proceedings, if they desire so.
8. The decision of JMC to this effect shall be final binding on the tenderer(s). The bid will be opened in front of the Committee, at the Office of the JMC.

3.4 Bid Preparation Cost

1. The Bidder is responsible for all costs incurred in connection with participation in this process, including, but not limited to, costs incurred in the conduct of informative and other diligence activities, participation in meetings/discussions/presentations, preparation of bid, in providing any additional information required by JMC to facilitate the evaluation process, and in negotiating a definitive Contract or all such activities related to the bid process. JMC will in no case be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.
2. This bid does not commit JMC to award a contract or to engage in negotiations. Further, no reimbursable cost may be incurred in anticipation of award. All materials submitted by the Bidder shall become the property of JMC and may be returned at its sole discretion.

3.5 Pre-Bid Meeting

Authority will host a Pre-Bid meeting as per the date mentioned in the Section 3.1. The representatives, limited to 2, of the interested organizations may attend the pre-bid conference at their own cost. The purpose of the conference is to provide bidders with information regarding the RFP and the proposed solution requirements in

reference to the RFP. Pre-Bid meeting will also provide each bidder with an opportunity to seek clarifications regarding any aspect of the RFP and the project.

All Bidders shall e-mail their queries to jmc.project27@gmail.com in the form and manner as prescribed in Annexure IX. The response to the queries will be published on 04-04-2025. No telephonic queries will be entertained thereafter. This response of Authority shall become integral part of RFP document.

3.6 Amendment of Bid Document

1. At any time before the deadline for submission of bids, JMC, may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective Bidder, modify the bid document by an amendment. All the amendments made in the document would be informed to all the participating bidders through email.
2. The bidders are advised to visit the JMC website as stated in this document on regular basis for checking necessary updates. JMC also reserves the rights to amend the dates mentioned in this bid for bid process. It will be assumed that the amendments have been taken into account by the Bidder in its bid.
3. In order to afford prospective Bidders reasonable time in which to take the amendment into account in preparing their bids, JMC may, at its discretion, extend the last date for the receipt of Bids.

3.7 Rights to Terminate the Process

1. JMC may terminate the bid process at any time and without assigning any reason. JMC makes no commitments, express or implied, that this process will result in a business transaction with anyone.
2. This bid document does not constitute selection offer by JMC. The Bidder's participation in this process may result in JMC selecting the Bidders to engage in further discussions and negotiations towards execution of a contract. The commencement of such negotiations does not, however, signify a commitment by JMC to execute a contract or to continue negotiations. JMC may terminate negotiations at any time without assigning any reason.

3.8 Earnest Money Deposit (EMD) and Refund

1. Bidders shall pay, along with their Bids, EMD amount as mentioned in Section 3.1. and it shall be paid online and should not be paying the EMD in personal.
2. In case a bid is submitted without the EMD as mentioned above then JMC reserves the right to reject the bid without providing opportunity for any further correspondence to the Bidder concerned.
3. The EMD shall be denominated in Indian Rupees only. No interest shall be paid by JMC towards the deposited EMD.
4. The EMD of the successful bidder will be discharged when the bidder has signed the agreement and furnished the required EMD. The EMD of L-2 and other higher bidders (L-3, L-4, etc.) shall be refunded immediately after opening of financial bid. In case, the successful bidder becomes non-responsive or successful bidder withdraws the bid or is unwilling to extend the bid validity bid validity period, in such circumstances, if L-2 bidder is agreeable to extend the bid validity period and ready to deposit the requisite amount of EMD to the department within the stipulated time period i.e., 15 days, the department will process further as per normal procedure.

The entire EMD may be forfeited in any of the following cases:

- If a Bidder withdraws its bid or increases its quoted prices during the period of bid validity or its extended period, if any
- In the case of all successful Bidders, if the Bidders fails to sign the Contract or to furnish Performance Bank Guarantee within specified time in accordance with the format given in this bid document
- During the bid process, if a Bidders indulges in any such deliberate act as would jeopardize or unnecessarily delay the process of bid evaluation and finalization
- During the bid process, if any information is found wrong / manipulated / hidden in the bid
- The decision of JMC regarding forfeiture of the EMD and rejection of bid shall be final and shall not be called upon question under any circumstances.
- EMD shall be returned to the unsuccessful bidder within 120 days from the date of opening of the Commercial bid.

- EMD of Successful bidder will be returned after the award of contract and submission of the performance Bank Guarantee within specified time.

3.9 Submission of Bid

1. Complete bidding process will be online (e-Tendering) in Three packet system. All the notification and details regarding terms and conditions related to this bid notice hereafter will be published online on web site <https://tender.nprocure.com/>
2. Bidding documents can be seen, downloaded, and submitted in electronic format on the JMC website. The deadline for submission of bid is specified of this document.
3. Technical bids will be opened as per the details found of this document.
4. Bidders should submit information and scanned copies of all the documents in PDF format in Pre-Qualification folder (Packet 'A') as mentioned in the Bid Document. Bidder should submit information and scanned copies pertaining to technical offer in PDF format in (Packet 'B') as mentioned in the Bid Document.
5. Bidder may be requested to submit original documents for verification during evaluation of technical bids as and when required.
6. Time and date of opening of financial bid will be informed by Email to technically qualified bidders.
7. An authorized representative of the Bidder should have valid class II / III Digital Signature Certificate (DSC) obtained from any Certifying Authority. The authorized representative of the Bidder shall digitally sign the original Technical bid and Commercial bid. The authorization shall be in the form of a written power of attorney accompanying the bid or in any other form demonstrating that the representative has been duly authorized to sign.
8. On opening the Pre-Qualification cover (Packet A) and Technical Cover (Packet B), if it is found that the Bidder has not submitted required documents as per Pre-qualification Criteria and Technical Criteria (Curable Defects) then the Bidder shall be intimated to comply with the said documents within 3 working days from the intimation from JMC (through email), otherwise 3% of the EMD paid by the bidder shall be forfeited and bid shall be treated as non-responsive.

9. JMC reserves the right to accept or reject any or all the Bids without assigning any reason. Moreover, if no intimation is provided by JMC then the documents submitted cannot be deemed as accepted.

3.10 Bid Submission Format

The entire bid shall strictly be submitted as per the formats specified in this bid.

3.11 Withdrawal of Bid

1. A Bidder wishing to withdraw its bid shall notify to JMC by e-mail prior to the deadline prescribed for bid submission. A withdrawal notice may also be sent by electronic means such as e-mail, but it must be followed by a signed confirmation copy, postmarked no later than the deadline for submission of bids. The notice of withdrawal shall be

- addressed to JMC at the address named in the Bid Data Sheet, and
- bear the Contract name, the <Title> and < Bid No.>, and the words “Bid Withdrawal Notice.” Bid withdrawal notices received after the bid submission deadline will be ignored, and the submitted bid will be deemed to be a valid submitted bid

2. No bid should be withdrawn in the interval between the bid submission deadline and the expiration of the bid validity period specified in the Bid Data Sheet.

Withdrawal of a bid during this interval may result in the forfeiture of the Bidder's EMD

3.12 Evaluation Process

1. The evaluation process of the bid proposed to be adopted by JMC is indicated in this section. The purpose of this section is to provide the Bidder an idea of the evaluation process that JMC may adopt.

2. JMC shall appoint a Bid Evaluation Committee (BEC) to scrutinize and evaluate the technical and commercial bids received. The BEC will examine the Bids to determine whether they are complete, responsive and whether the bid format conforms to the bid requirements. JMC may waive any informality or non-conformity in a bid which does not constitute a material deviation according to JMC.

3. There should be no mention of bid prices in any part of the bid other than the Commercial Bid
4. Any attempt by a bidder to influence the bid evaluation process may result in the rejection of Bid.

3.13 Evaluation of Technical Bid

1. The Technical Bids of only those Bidders, who qualify in the Pre-Qualification stage, shall be considered and will be evaluated as per the evaluation criteria in this clause. The Bid Evaluation Committee (BEC) shall invite each Bidder to make a presentation-cum demonstration as part of the technical evaluation.
2. JMC shall appoint a Bid Evaluation Committee (BEC) to scrutinize and evaluate the technical and commercial bids received. The BEC will examine the Bids to determine whether they are complete, responsive and whether the bid format conforms to the bid requirements. JMC may waive any informality or non-conformity in a bid which does not constitute a material deviation according to JMC.
3. There should be no mention of bid prices in any part of the bid other than the Commercial Bid. Presence of details regarding commercials in any other section or document would lead to disqualification of the bidder.
4. ~~Bidders should not submit physical copies of the bid to the . Submission of the same would lead to disqualification of the bidder.~~
5. The BEC may require written clarifications from the Bidders to clarify ambiguities and uncertainties arising out of the evaluation of the Bid documents (to be stated precisely as it should be in JMC's interest). In order to qualify technically, a Bid must secure a minimum of 70% of total marks in technical evaluation after summing up. Only those Bids which have a minimum score of 70% of total marks in technical evaluation will be considered for opening of their Commercial Bid. However, the BEC reserves the right to lower the minimum required marks if none of the Bidders achieves 70% of the total marks. Only the Bids qualifying the technical evaluation will be considered for commercial evaluation.
6. The bidders' technical solutions proposed in the bid document will be evaluated as per the requirements specified in the Bid document and adopting the evaluation criteria spelt out is below.

3.14 Technical Criteria

#	Evaluation Criteria	Maximum Marks	Criteria	Supporting Document
A1	Financial Capability of the Sole bidder	10	Average Annual Turnover of Sole Bidder from GIS services for the last three financial years: • >= □ 25Crore – 5 marks • >= □ 30Crore - 10 marks	Copy of audited Balance sheet along with Chartered Accountant clearly stating the average annual turnover GIS services over three financial years.
B1	Bidder's Competence for execution	5	The Sole Bidder must have executed ULB / Smart Cityprojects containing 2D mapping for the cities in last five financial years with minimum project value of 1.5 Cr: 1. 2 Project –2 Marks 2. 3 Project – 3 Marks 2. 4 or More Projects - 5 Marks	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope
B2	Bidder's Competence for execution	10	The Sole Bidder should have developed 2D & 3D Web GIS platform for ULB / Smart Cityprojects in last five years • 2 Project –2.5 Marks • More than 2 Projects – 5Marks • Additional 5 Marks for experience in 2D & 3D web	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope

			solution on COTS Platform	
B3	Bidder's Competence for execution	15	COTS platform to be Make in India (More than 50%) Make in India COTS GIS Platform – 10 marks Additional 5 marks if full IPR and source code is within Indian boundaries	
B5	Bidder's Competence for execution	15	The Sole Bidder should have experience in ICCC integration with GIS for Central / State Government / Cooperative / ULB/ Smart City project in India of overall project of value not less than INR 1.5crore each during the last 5 years: 1 project - 7marks 2 project – 10marks More than 2 project - 15marks	1. Valid Work Order/PO issued by the client 2. Go-Live/Completion Certificate issued by the client 3. The certificates should be issued by the competent authority clearly stating the scope
B6	Certifications	5	ISO 9001:2015 – 1 Mark ISO 27001:2022– 2Mark CMMi Level 3 Dev or above – 2Mark	Copy of valid certificate/s to be provided
B7	Bidder's Competence	5	Project Manager - PMP Certification with Post	1. Proof - PMP Certification

	e for execution		Graduation /M.Sc/M.Tech background having 15+ Years Experience in GIS Projects– 5 marks – 5 marks	
C1	Technical Proposal	5	Quality of Technical Proposal: - Overall approach of the bidder for implementation of change detection technique and Web Geo spatial system, Integration with existing GIS applications and data mentioned in this RFP	Technical Proposal
D1	Presentation of Proposal	20	Presentation of Change Detection of City model created/ delivered by the bidder and its streaming on web Presentation with following functions: Geo enabled Dashboard Function to view the network	Presentation to be given to MC as per the schedule provided by JMC.
D2	Proof of Concept/ Demonstration	10	•Demonstration on Change Detection of the Urban landscape - 5 Marks •Demonstration on Analytics on web portal– 5 marks	POC/demonstration to be given to MC as per the schedule provided by JMC.
Total marks			100	
Note: -Minimum marks required to qualify-70				

3.15 Opening of Technical Bid

1. JMC shall open the Technical Bids in public, in the presence of Bidders' designated representatives and anyone who chooses to attend, at the address, and at the date and time specified.

2. Only bids that are opened and read out at the bid opening and whose EMD has been paid online through the e-Tendering Application shall be considered further.

3. All the bids shall be opened one at a time, reading out the name of the Bidder, the presence of an EMD, and any other details as JMC may consider appropriate.

4. JMC shall prepare a record of the bid opening that shall include, at a minimum: the name of the Bidder and the presence or absence of EMD. The Bidders' representatives who are present shall be requested to sign the attendance sheet.

5. Authorization letter in the name of the person attending bid opening meeting needs to be submitted on the letterhead of the Bidder during bid opening in the format specified in Annexure VI.

6. Once the bids are opened each bid will be checked for pre-qualification criteria.

1. The financial bids shall not be opened by JMC until the evaluations of the Pre-qualification and Technical Bids have been completed.

2. After the technical evaluation is completed and JMC has issued its no objection (if applicable), JMC shall notify those Bidders whose proposals did not pass the technical evaluation or were considered as non-responsive to the Bid Document and scope of work, that their Financial Proposals will not be opened.

3. JMC shall simultaneously notify in writing to bidders who have cleared the technical evaluation, the date, time and location for opening the Financial Proposals. The opening date would allow Bidders enough time to make arrangements for attending the opening. Bidders' attendance at the opening of Financial Proposals is optional.

4. JMC shall prepare a record of the bid opening that shall include, at a minimum: the name of the Bidder and the bid Price, and any other details as JMC may consider appropriate. The Bidders' representatives who are present shall be requested to sign the attendance sheet. A copy of the record shall be distributed to all Bidders.

5. Financial Proposals shall be opened publicly in the presence of the Technically Qualified Bidders' representatives who choose to attend. The name of the Technically Qualified Bidders shall be read aloud.
6. Commercial (Financial) Bids from bidders who have failed to qualify in evaluation of the technical bid will not be opened.
7. Only bids that are opened and read out at the bid opening shall be considered further.
8. Authorization letter in the name of the person attending bid opening needs to be submitted on the letterhead of the Bidder during bid opening in the format specified in the bid document.
9. The Commercial Bids will be evaluated by JMC for completeness and accuracy.
10. Activities and items described in the Technical Proposal but not priced, shall be assumed to be included in the prices of other activities or items.
11. Based on the technical evaluation criteria, only those bidders scoring 70% in the technical evaluation will be shortlisted for commercial evaluation.

3.16 Selection Method

1. Based on the technical evaluation criteria, only those bidders scoring 70% in the technical evaluation will be shortlisted for commercial evaluation.
2. The Technical Score of the bid under consideration will be noted as S_t where it represents the net marks scored in the technical evaluation by the bid under consideration.
3. The formula for determining the financial scores is as follows: $S_f = 100 \times F_m/F$ in which S_f is the financial score of the bid under consideration, F_m is the lowest price across all received bids and F is the price quoted for the bid under consideration.
4. The weights given to the Technical and Financial aspects are 0.8 and 0.2
5. The formula for determining the Net Score (N_s) of each bid under consideration would be $N_s = (S_t \times 0.8) + (S_f \times 0.2)$ where S_t and S_f are the technical and financial scores of the respective bids
6. The bidder attaining the highest Net Score, i.e. N_s would be awarded the contract.

3.17 Rights to Accept/Reject any or all Bids

Notwithstanding anything contained in this TENDER, JMC reserves the right to accept or reject any Application and to annul the Bidding Process and reject all Applications / Bids, at any time without any liability or any obligation for such acceptance, rejection or annulment, and without assigning any reasons therefore. In the event that the Authority rejects or annuls all the Bids, it may, in its discretion, invite all eligible Bidders to submit fresh Bids hereunder.

JMC reserves the right to reject any Application and / or Bid if:

- (a) At any time, a misrepresentation is made or uncovered, or
- (b) The Applicant does not provide, within the time specified by the Authority, the supplemental information sought by the Authority for evaluation of the Application. In case it is found during the evaluation or at any time before signing of the Agreement or after its execution and during the period of subsistence thereof including the concession thereby granted by JMC, that one or more of the pre-qualification conditions have not been met by the Bidder, or the Bidder has made any misrepresentation or has given any incorrect or false information, the Bidder shall be disqualified forthwith if not yet appointed as the Successful Bidder(s) either by issue of the LOA (Letter of Approval/Acceptance) or entering into of the Agreement, and if the Bidder(s) has already been issued the LOA or has entered into the Concession Agreement, as the case may be, the same shall, notwithstanding anything to the contrary contained therein or in this TENDER, be liable to be terminated, by a communication in writing by JMC to the Bidder, without Bidder being liable in any manner whatsoever to the Bidder and without prejudice to any other right or remedy which JMC may have under this TENDER, the Bidding Documents, the Concession Agreement or under applicable law.

JMC reserves the right to verify all statements, information and documents submitted by the Bidder in response to the TENDER. Any such verification or lack of such verification by JMC shall not relieve Bidder of its obligations or liabilities hereunder nor will it affect any rights of JMC there under.

The bid shall be rejected if the bidder

- a. Stipulates the validity period less than 180 days.

- b. Stipulates own condition/conditions.
- c. Does not fill and (digital)* sign undertaking forms, which are incorporated, in the document.

*- Using Digital Signature Certificate (DSC) class II / III

3.18 Clarification

Bidder requiring any clarification on the tender may notify JMC in writing or e-mail. They should send in their queries before the date specified. JMC shall Endeavour to respond to the queries within the period specified therein. The responses will be sent by Email. JMC will forward all the queries and its responses thereto, to all purchasers of the TENDER without identifying the source of queries.

JMC shall Endeavor to respond to the questions raised or clarifications sought by the Bidder. However, the JMC reserves the right not to respond to any question or provide any clarification, in its sole discretion, and nothing in this Clause shall be taken or read as compelling or requiring the JMC to respond to any question or to provide any clarification, but not later than the date provided.

JMC may also on its own motion, if deemed necessary, issue interpretations and clarifications to all Bidders. All clarifications and interpretations issued by JMC shall be deemed to be part of the tender. Verbal clarifications and information given by JMC or its employees or representatives shall not in any way or manner be binding on the JMC

3.19 Amendment of Bid Document

At any time prior to the deadline for submission of Application, JMC may, for any reason, whether at its own initiative or in response to clarifications requested by a Bidder, modify the tender by the issuance of Addendum.

Any Addendum thus issued will be published on web site or sent in writing or Email to all those who have purchased the tender.

In order to afford the Bidder a reasonable time for taking an Addendum into account, or for any other reason, JMC may, in its sole discretion, extend the Application Due Date

3.20 Notifications of awards (LOA) and Signing of Contract

1. Prior to the expiration of the period of bid validity, the Bidder will be notified in writing or by Email that their bid has been accepted.
2. At the time JMC notifies the successful Bidder that their bid has been accepted, JMC will send the Bidders the proforma for Contract, incorporating all clauses/agreements between the parties. The successful Bidder(s) shall sign and date the Contract and return it to JMC. Draft Format of the contract has been included in the bid document. Final Contract agreement will be provided after due verification from JMC's legal department.
3. Penalty of Rs.1000/- per day will be applicable in case of no submission of contract agreement within stipulated period.
4. In case of pending execution of contract agreement, Payment of 3% of Total contract value will be withheld.

3.21 Performance Bank Guarantee

1. The Bidder shall at his own expense, deposit with Corporation, within thirty (30) working days of the notification of award of the contract an unconditional and irrevocable Performance Bank Guarantee (PBG) from the list of approved banks (specified in the bid document) as per the format given in this bid, payable on demand, for the due performance and fulfillment of the contract by the Bidder.
2. This Performance Bank Guarantee will be for an amount equivalent to 3% of the total contract value. All charges whatsoever such as premium, commission, stamp duty etc. with respect to the Performance Bank Guarantee shall be borne by the Bidder.
3. Details of validity of the performance bank guarantee are specified under section 3.1 of this document. The Performance Bank Guarantee letter format can be found in this document.
4. The Performance Bank Guarantee may be discharged / returned by Corporation upon being satisfied that there has been due performance of the obligations of the Bidder under the contract. However, no interest shall be paid on the Performance Bank Guarantee.

5. In the event of the Bidder being unable to service the contract for whatever reason, JMC would evoke the PBG. Notwithstanding and without prejudice to any rights whatsoever of JMC under the Contract in the matter, the proceeds of the PBG shall be payable to JMC as compensation for any loss resulting from the Bidder's failure to complete its obligations under the Contract. JMC shall notify the Bidder in writing of the exercise of its right to receive such compensation within 14 days, indicating the contractual obligation(s) for which the Bidder is in default.

6. The 30 days' notice period shall be considered as the 'Cure Period' to facilitate the System Integrator to cure the breach. The PBG shall be evoked only if the breach is solely attributable to the bidder and the bidder fails to rectify the breach within the 'Cure Period'.

7. JMC shall also be entitled to make recoveries from the Bidder's bills, performance bank guarantee, or from any other amount due to the Bidder, the equivalent value of any payment made to the Bidder due to inadvertence, error, collusion, misconstruction or misstatement.

3.22 Failure to agree with the Terms and Conditions of the Bid/Contract

Failure of the Bidder to agree with the Terms and Conditions of the bid/Contract shall constitute sufficient grounds for the annulment of the award of contract, in which event the contract may be awarded to the next most responsive Bidder which means the bidders who has score second highest as per evaluation method. In such a case, JMC shall invoke the PBG of the most responsive Bidder.

3.23 Terms and Conditions of the Bid

Bidder is required to refer to the draft Contract Agreement, provided in this bid, for all the terms and conditions (including project timelines) to be adhered by the successful Bidder(s) during contract period. Please note that one needs to read the Contract Agreement as a whole document.

GENERAL CONDITIONS OF CONTRACT

Responsibility of the Bidder

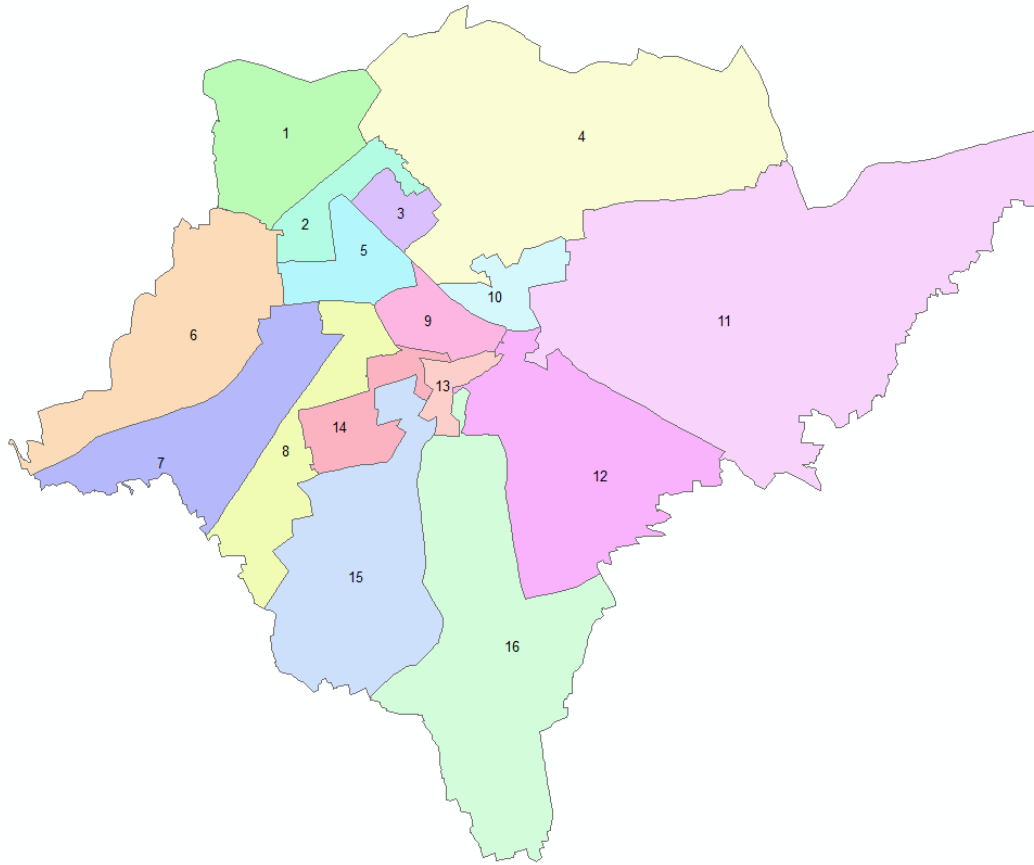
- a) Bidder must appoint a Project Manager, directly accessible to JMC, to be responsible for the entire project including matters such as personnel management, contracts administration, progress monitoring, logistic support, documentation, and operational set-up. He/she shall have the authority to

make decision concerning the execution of all contractual activities & to accept & act upon any JMC decision & be responsible for the smooth implementation with the help of specialized experts working under his/her control for the whole project.

- b) Bidder must submit the detailed schedule of implementation for various activities to be performed under the contract & shall include the information required for JMC date wise. Bidder shall submit overall monthly progress & status report to JMC. In addition to that the Project Manager shall give report of the progress of the work up to date & the plans for the accomplishment of work remaining to be performed in the progress review meeting as & when required.
- c) The bidder shall also make multimedia presentations to the officers of JMC for clearing the concept of various technology / systems to be developed / systems developed / etc. as and when it is seems necessary by the JMC so as to create greater awareness for computerization and greater acceptance of developed systems. The hardware and other necessary material for the same shall be provided by JMC.
- d) Bidder shall not disclose/submit/supply/give the Map/attribute/database information that they shall access during the phase of preparation & implementation of this work to any third party/person. A Non-Disclosure agreement shall be signed by bidder.
- e) Till the validation of customized software by JMC all development work shall be carried out on Bidder's own hardware & software at their own premises.
- f) Bidder shall have an office physically located within JMC jurisdiction limits. The office shall be equipped with all necessary telecommunication like telephone, fax, internet, etc. and other hardware and software requirements for carrying out day to day work on the JMC's project/application.
- g) It is likely that JMC may extend the work to surrounding developed areas and bidder shall be liable to perform the work on mutually acceptable terms and conditions.

4 Scope of Work

4.1 Area of Interest



Jamnagar Municipal Corporation is approximately 150 sq. Km and there are 16 wards in Jamnagar.

4.2 Requirement Gathering

The requirement gathering is the phase where SI is expected to get maximum information, data & status of existing system from JMC user departments. The SI is expected to deploy Project Manager & Business Analysts to meet different stakeholders & JMC user departments. These two resources are expected to attend all meetings related to this project. These resources are expected to brief the back-end development & implementation team of client's requirements, expectations & decisions made with respect to the project.

4.2.1 FRS, SRS, and Project Plan

The System Integrator should carry out detailed requirement gathering with IT Department and other concerned departments to detail out all the requirements in Functional Requirement Specifications (FRS) towards creation of Software Requirement Specifications (SRS) and Standard Operating Procedures (SOP) to be designed for the proposed solution. The System Integrator should study and collect the existing spatial and non-spatial datasets as per the requirements furnished in the scope. The SI needs to prepare a project plan as per their understanding, interaction with JMC stakeholders & their experience in delivering projects.

4.2.2 Project Kick-Off Meeting & Presentation to Key Departments of JMC

The SI should conduct the project kick-off meeting at the start of the project to understand the key stake holders, client's expectations, and other necessary details about the project & JMC stakeholders. The SI should conduct presentation to key departments of the JMC.

4.3 Successful Bidder/System Integrator is supposed to:

- A. Procure Satellite Images (archival-based on availability& fresh tasking) of different time – (1990-2024)10 sets
- B. Creation of 2DBase Map of Jamnagar City by using Satellite Imagery complying AMRUT guidelines.
- C. Collection of Ground control points (GCP) to geo-reference Satellite images.
- D. Apply AIML, Change Detection using Satellite Imagery for end user selected 2 sets of Imagery
- E. Survey of underground utilities for JMC
- F. Design and development of Mobile application for property assessment and Change Detection to validate and collect/update details of properties which are identified as changes.
- G. Design and development of COTS based Spatially enabled Web GIS Decision Support System(DSS) 2D and 3D compatible with Geo-enabled Dashboards to

incorporate the JMC department functional data with visualization tool to display attribute and spatial data

- H. Design and development of Spatially enabled Web Citizen portal for data access with citizen centric modules as defined by JMC
- I. Hosting of data on JMC storage.
- J. Training and Capacity Building Programs for Department Officials.
- K. Setting up of GIS Cell
- L. Operation & Maintenance for the period of 3years.

4.3.1 Procurement of Satellite Image

1. The successful bidder shall procure Satellite imagery (30-50 CM) of AOI for yearly basis (archived imagery for last ten previous years from year 2024)
2. Satellite imagery should be of latest or 8 months old from the issuance of LOI/Work order.

4.3.2 Differential Global Positioning System(DGPS) Survey for Collection of Ground Control Points (GCPs)

1. Establishing of Ground Control Points for Geo-referencing of satellite imagery.
2. Identification and marking of Ground Control Points (GCP) on the area of Interest.
3. At least 1 Ground Control Point need to be established on ground for 1 Sq.KM. The GCP can be temporary or permanent. It should be taken with double frequency DGPS (Digital Global Positioning System).
4. Report shall be prepared choosing Ground Control Points with easy access as well as easy identification on satellite imagery.
5. Control points shall be established on ground based on the finalize locations preferably on permanent structure, culvert corner, Road barrier, bridge corner, rock-in-situ, field bunds and other definable points like tri junctions, road-junctions.

6. A detailed survey report shall be prepared consisting of GCPs parameters and pictures for JMC validation and records

4.3.3 2D Base map generation using satellite imagery

1. The bidder should prepare a detailed, up to date 2D Base map for the latest acquired satellite Imagery. JMC shall provide the available base layers to the selected bidder. The bidder shall prepare a 2D base map of all the wards/zones coming under JMC authority.
2. Data Model for storing the spatial & Non-Spatial data shall be decided in consultation with the JMC. Need to finalize data model for creating/populating data in base map as well as providing intended outcome of GIS enables web applications.
3. Every layer should have attribute information such as GIS ID and Other Attribute information shared by JMC and complying AMRUT Guidelines.
4. The 2D data output shall be compatible across industry standard formats and comply with AMRUT Guidelines
5. The selected bidder shall create 2D Base Map of the identified JMC Project Area. Plotting/feature extraction/vector compilation shall be done at 1:1000 scale.
6. The Successful bidder shall also provide 2D city data of every year for performing change detection analysis to visualize the growth of the city for the next 3 years.
3. 2D Base-map should be free from errors related to digitization.
4. The base map will be prepared in various layers for ease of operation in GIS application. The details of the various layers to be part of the final base map are given as below, which should be prepared using image and existing data available with Department. All other layers as per the AMRUT Guidelines and JMC requirement

Sr No	Layer Name	Geometry Type	Data Source
1	Municipal Boundary	Polygon	JMC
2	Zone Boundary	Polygon	JMC
3	Ward Boundary	Polygon	JMC
4	Road Network	Line	Imagery
5	Building Footprints	Polygon	Imagery
6	Streetlights	Point	JMC
7	Bridges/Flyover	Polygon	Imagery
8	Water Bodies	Polygon	Imagery
9	Railway Network	Line	Imagery
10	Footpath & Divider	Polygon	Imagery
11	Municipal Corporation Buildings	Point	JMC
12	Historical Buildings	Polygon	JMC
13	Parks and Garden	Polygon	Imagery
14	Smart Components	Point/Line/Polygon	JMC

4.3.4 LULC (Land Use Land Covered) Map

- LULC maps to be created as per the availability of the images. LULC maps can include Urban Clusters, Barren Land, Shadows, Water bodies, Rivers, Nalas, Forest, Agricultural Land, and other landmarks.
- The bidder shall be able to create LULC maps using Supervised, Unsupervised, as well as Object Oriented Classification algorithms provided as a part of software module. It will bring about classification of man-made and natural features.

4.3.5 Survey of Underground Utilities: -

Under Ground Utility Mapping for the City is intended to capture information about following sub surface utilities below 5 meter depth from ground for pilot area:

- Water Networks – Including all Pipe Lines, Valves, Underground storage tanks, Storm network.
- Sewerage Networks – Including man holes, treatment plants, and booster pumps
- Storm water Networks – Including man holes

Note: Based on the results of POC from the pilot area of 10 KM, JMC reserves the right to increase the quantity of survey of the quoted price for the KM's as needed

4.3.5.1.1 Detailed scope and activities to be undertaken are given below

- ✓ The Successful Bidder shall locate and identify defined underground services using Ground Penetration Radar (GPR) Technology within JMC Jurisdiction as per direction of engineer in-charge
- ✓ Underground utilities, associated surface features, change of direction and bifurcation shall be located and X, Y with depth needs to be recorded at intervals not exceeding 50 meters.
- ✓ The estimated length of the GPR survey for utilities shall be around 1200 KM's length. However, length of the survey could be increased or decreased during the survey activities. Overall, length of the line survey shall be considered on actual length of the length surveyed
- ✓ At any significant change (more than 0.50m) of depth below the ground, an annotation shall be provided. This annotation shall be placed at the same Z – value (depth level) as recorded at that very point. Each service shall be annotated with the type of utility, depth, and diameter of pipe at appropriate intervals
- ✓ The Successful bidder shall be responsible for co-ordination with various city agencies and State Government Departments & utilities whenever and wherever necessary on the behalf of JMC for the purpose of survey.
- ✓ Utilities visible on surface, such as drains, manholes, chambers and draw pits shall be located and mapped to the accuracy as specified mapping accuracy.
+/- 25 CM against 1-2.5 Meter Depth of the utilities
+/-50 CM against to 2.5-5 Meter Depth of the utilities

- ✓ Underground utilities shall be mapped continuously and recorded in three dimensions ensuring that sub surface features are captured at each change of direction and bifurcation
- ✓ Positions and levels shall be related to the specified grid and datum and shall normally be related to the center of metallic pipes or cables, crown of ducts and inverts of sewers and storm water drains
- ✓ Any known underground utilities or information which cannot be mapped to the accuracies stated under mapping accuracy other than by excavation, shall be entered in a unique layer defined as “uncharted,” as shall be approved by JMC.
- ✓ The Vendor shall itemize in his reports the types of utilities which have been classified as “uncharted” and other circumstances, such as local areas of interference, where the specified accuracies cannot be achieved.
- ✓ Undulations in land surface during utility survey to be taken care off
- ✓ As part of input data/material to conduct survey, JMC shall provide following to the Successful Bidder (if available);
 - Utility Drawings (Hard/Soft Copy) of the utilities to be mapped
 - Available GIS data / base maps already generated, if any
- ✓ The Successful Bidder shall take all-possible precautions while carrying out such survey and shall ensure not to create any traffic inconvenience, as well inconvenience to the Public at large and avoid any survey to be carried out in restricted areas, if any as per JMC directions.
- ✓ The survey and record making shall be carried out as per guidelines and technology mentioned above. In case of any other methodology is to be used by the successful bidder, the same may be agreed by JMC for specific areas where there are accessibility issues, provided end results are same in terms of recording the network details spatially in the target system i.e. Web application.
 - Each Utility should have minimum following attribute data attached with it: -
 - Type of Utility (Water, sewerage, Storm)
 - Depth of Utility in meters.
 - Size (Diameter and length) of utility in meters.
 - Type of material

4.3.6 AI/ ML based Land Change Detections

The system will use cutting-edge technology such as artificial intelligence and machine learning (AI/ML) and mobile applications to assist the Jamnagar Municipal Corporation in monitoring and safeguarding layouts, unauthorized building constructions, deviations, and encroachment. Further, the ML based change detection system should have the following features/requirements

- The ML system must be adaptable so that, custom models developed must be able to accommodate.
- The system must have model inventory dashboard in which user should be allowed to visualize side-by-side comparison of models based on performance, and prediction outcomes.
- Interactive visualization of land use change alerts, their status, customized reports for authorities and user specific actionable dashboard should be available under analytics.

4.3.6.1 Satellite based Layout Monitoring

The **Satellite-Based Layout Monitoring Module** is envisioned as a critical component for the real-time monitoring of land encroachments and unauthorized constructions within the jurisdiction of Jamnagar city using advanced satellite imagery, Geographic Information Systems (GIS), and data science technologies. The module shall be capable of detecting various types of land use, including new constructions, under-construction sites, and other man-made structures, through the application of geo-referenced and ortho-rectified satellite data, along with land deviation models.

Vintage	Image Resolution	Deliverables
1990-2023 (9 Sets)	As per the availability	Ortho-mosaic map, LULC Map, Change Detection Map year-on-year
2024	30-60 cm (As per the Availability)	Ortho-mosaic map, LULC Map, Vector model of buildings, slums, major infrastructure like Metro, Bridges, Flyover & Base-map

2025-2029 (5 Sets)	30-60 cm (As per the Availability)	Ortho-mosaic map, LULC Map, Change Detection Map year-on-year
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The system shall provide the following key functionalities:

- **Visualization of Land Use Changes:**

The system shall enable the visualization of land use changes, particularly in unapproved layouts and unauthorized constructions on both government and private land parcels. The solution shall assist in detecting and highlighting areas where unauthorized development has occurred or is ongoing.

- **Integration with the Building Permission Portal:**

The system shall be integrated with the existing online building permission portal through the use of APIs, web services, or other mechanisms. The integration shall provide near real-time access to building permission statuses for various zones or areas, along with relevant analytics. This integration shall allow authorities to monitor ongoing constructions in relation to approved permissions.

- **Real-Time Updates of New Permissions:**

Any new building permission granted in a particular area shall be automatically updated within the system. The system shall ensure the data is refreshed in near real-time, providing up-to-date information on the current status of building permissions.

- **Automated Detection of Deviations:**

The solution shall utilize data science techniques, such as machine learning and artificial intelligence, to detect deviations in land use pattern and change in any existing or new developments against yearly satellite imagery.

4.3.6.2 Reports & Analytics

The System shall provide informational insights in the form of Spatial and MIS dashboards and reports. The Spatial/MIS dashboard shall be able to provide stakeholders with easy-to-understand, easy-to-use reports that use appropriate infographics (Charts), filters and download options to monitor the performance.

Following reports to be made available in downloadable xls/pdf format:

- i. Change in Land use pattern for existing or new construction for validation by JMC

iii. Ward/zone/city wise verification report

iv. Monthly / annual alerts report

4.3.6.3 Mobile Application for Change Detection Validation

- All the mobile input data should be collated from the users entering different types of data from across the admin boundaries and assimilated in the appropriate database.
- The frequency of updating the data input from the mobile can be anything from every few minutes to everyday depending on the source and type of the data.
- The system should be designed to accommodate any frequency of data submissions and inputs.
- Alerts for the officials should contain the land parcel details with latitudes and longitudes and land-use change related satellite image.
- Mobile App should support both iOS and Android operating Systems.
- It shall support the following workflows:
 - Land-use change alerts generated by satellite imagery shall be automatically available in Mobile Apps based on the user Profiles as per the jurisdictions.
 - Land parcel details with latitudes and longitudes, as well as land-use change related satellite image, shall be included in alerts for officials.
 - Department officials shall be able to easily navigate to hotspots using the Mobile App and report their actions using Texts, Geotagged Photos, and Videos on the Mobile App.
 - The mobile app shall display and allow the user to update the status of identified hotspots based on the actions listed below.
 - ✓ Identified
 - ✓ Validated
 - ✓ Issues Raised
 - ✓ Resolved

4.3.7 Development of COTS based Enterprise Web Based Decision Support System(DSS) with Operational Dashboards

ACOTS based Enterprise level GIS is a system that is integrated through an entire organization so that multiple users can manage, share, use spatial data and related information to acknowledge and work on varied requirements such as data creation, data updating, data visualization and analysis and data/information dissemination.

An Enterprise GIS, which will be 2D and 3D compatible, aims at providing an integrated and interoperable system environment where multiple functions of the same enterprise can work together and at the same time can take care of external collaborations for betterment their core functions individually.

Through this project, it is expected that the bidder will design and develop an Enterprise GIS system for the Jamnagar Municipal Corporation along with the data and maps mentioned above, following are the key components that will constitute to an Enterprise level GIS. JMC envisions to utilize this platform to consume 3D data without any extra cost or efforts to be put in when the 3D data is built by JMC

The selected bidder should develop a web-based GIS application/portal for JMC. This application will primarily use in viewing, analyzing, & utilizing the image repository in order to find out the changes happening on ground over the period. Web based GIS application shall have capability to utilize the change detection map & basic change detection analysis. In short, this system will play a role of Decision Support System (DSS) for JMC including Property/Revenue Tax Department.

4.3.7.1 Functional Requirements of Web Application for Department Portal

- Platform for GIS Application Software should be on COTS
- GIS Software must allow JMC to implement a centrally managed GIS providing the advantage of lower cost of ownership through single, centrally managed, focused GIS applications (such as a Web application)
- Proposed customized GIS Application Software should be able to integrate with the other Solution being developed at JMC
- Software should provide for audit trail of changes to the data
- The software should provide facility to click on any feature of the map and return a select set of attributes for feature

- Facility to perform the spatial intersection analysis like plot area with buffer zone to calculate road-widening impact on adjacent land.
- Software should support printing spatial data at different scales and at adjustable print quality
- The system should provide facility to exchange the GIS Data with other platform applications like Microsoft Word, and Excel to use GIS data and generate reports like graph and charts
- The software should allow user to open raster images, or satellite images of various standard format
- Software should have various query tools for queries based on attributes, location.
- The software should have a provision of hyper linking the GIS feature as well as its attribute fields with existing documents, drawing files or scanned maps related to that feature
- The software should have the ability to add data from internet or intranet users to the existing map data so that data from other sources can also be utilized by JMC
- Software should have basic Navigation tools like the software should have tools to Pan, Zoom, and Rotate the Map according to user requirements and should also be able to use Mouse Wheel for zoom-in and zoom-out, The software should have ability to search and to zoom into the user specified x, y coordinates, The software should have facility to create and organize user desired number of Spatial Bookmarks and should be able to share the same
- The software should support Map Services, Open Geospatial Consortium, Inc. (OGC), Web Coverage Service (WCS), Open Geospatial Consortium, Inc., Web Map Service (WMS), creation & consumption both.
- The software should have tools and options like raster management, conversion tools, proximity, analysis, packaging, spatial statistics, geometric network tools, data indexing

The following are the expected functional requirements

Web Base GIS Application/platform: The basic web-based GIS application will have following things:

➤ **Dashboard:**

- ✓ The basic dashboard will have departmental GIS module links which will be added as a part of post implementations phase. Also there will be a common GIS module on the dashboard where below mentioned generic GIS functionalities will be available.

➤ **Layer Management:**

- ✓ The solution available enterprise wide will be based and shall have capacity to manage layers by switching on or off layers. Also, transparency and visibility scale shall be fixed for each layer.

➤ **Attribute Viewer:**

- ✓ The Attribute Viewer allows you to view the attribute fields of selected features

➤ **Query Panel:**

- ✓ Select features in a layer using a SQL-based expression against their attributes

➤ **Predefined queries:**

- ✓ Predefined query to make it easier for users to view subsets of a map's data from an end-user perspective, executing the query is simple and is performed with a single button click. It is designed to work on a single layer.

➤ **Map Contents:**

- ✓ Map contents list all the layers on the map and shows what the features in each layer represent. The check box next to each layer indicates whether its display is currently turned on or off.

➤ **Buffer:**

- ✓ It creates a zone around a map feature (Point & Line) measured in units of distance or time. This buffer is useful for proximity analysis.

➤ **Proximity:**

- ✓ Selecting the features based on their location relative to features in another layer. For instance, if you want to know how many homes were falling in

certain Zone for tax implementation, you can select all the properties that fall within the Zone boundary.

➤ **Navigation**

- ✓ Pan View, Refresh Map
- ✓ Zoom to full extents
- ✓ Zoom In (User-defined area)
- ✓ Zoom Out (User-defined area)
- ✓ Zoom to Previous extents (Back)
- ✓ Zoom to Next extents (Forward)
- ✓ Zoom to Selected Features
- ✓ Activity indicator - display notification while map/data is being processed
Move up/down/left/right.
- ✓ Scale input box - allow user to enter representative fraction scale for dynamic services - For cached services, scale box should contain dropdown menu of available cache scales (levels of detail)
- ✓ Show/hide mouse coordinates

➤ **Descriptive Map Information Tool**

- ✓ On Clicking or Mouse Hovering over each map feature, information should be shown based on the feature's attributes. Functionality should be available for all feature classes; should be able to display a combination of attributes and should not limit the number of features that can be included with the map tool.
- ✓ It should allow user to turn on and off this attribute information as needed

➤ **Searching Tools**

- ✓ Should allow users to search features by both pre-configured and dynamic based on Search by Zone, Search by Ward Search by Parcel ID.
- ✓ Should allow user to run the custom queries on-the-fly and save those queries for future use
- ✓ Should allow user to run spatial query on multiple layers with spatial operators
- ✓ Should also allow for a buffer to be applied to the search criteria allowing for features within a certain distance of the query feature to be selected.
- ✓ Should have a tool allowing users to search and zoom to a location based on GPS coordinates
- ✓ Should have facility to run combination of attribute & spatial query

➤ **Distance and Area Measurements**

- ✓ Should have distance measurements tool to allow user to measure the length of irregular shaped lines
- ✓ Should have area measurements tool to allow user to measure irregular shaped polygons
- ✓ Measurements should be shown using the metric system.
- **Visualization of 3D Data**
 - ✓ The proposed platform shall be capable of rendering 3-dimensional GIS models of ground terrain, vegetation, water-bodies as well as man-made features.
 - ✓ This platform shall be available for access over the internet. The rendering shall be optimized for network bandwidth as well as high performance at client side with the help of thin client application.
- **14.Printing**
 - ✓ Should have ability to print maps to a printer/plotter with the selection of paper size (A2, A1, A0, Letter, Tabloid.) and page orientation (landscape or portrait)
 - ✓ Should have print preview option
 - ✓ Should be able to handle and process any redlining / markups of the map.
 - ✓ Should have ability to export the map to a standard image format (BMP, TIF, JPEG and PDF file)
 - ✓ Print date and time should be automatically added to output at application runtime
 - ✓ Legend should be automatically adjusted based layers displayed in print area
 - ✓ User will have option to choose from various features like scale style, legends. to be printed on the map (hard copy)

4.3.7.2 Functionalities of Citizen Portal

The bidder is expected to follow the complete SDLC for the development of the Web-GIS applications.

- **My Location:**

To locate citizens' current location, display nearby utilities & distance between current location and searched utility

- **Ward Information:**

Ward details, zone details, officers, educational institutes, health facilities, hotel, entertainment, grievances

- **Search:**

Searching services or utilities like Education, Emergency, Government offices, Health service.

- **Know your property:**

Searching one's property by property ID, phone number, address

- **Query:**

It will have common queries for citizens: Locate Toilet, Locate building based on construction type, Locate building by total number of floors, Locate Garbage disposal bins, Locate building by Usage type, Locate building by no. of floors, Locate slums, Locate hotels/bus stop/ATMs.

- **Draw:**

The tool should basically allow citizen to draw the shapes like Point, Line, Polygon, Triangle, extent, freehand polyline, circle, ellipse, freehand & save & share it in .pdf, .jpg or other format

- **Event:**

It displays the event in the city for next 7 days/1 month along with the details/phone numbers & coordinates, provision for the citizen to request to add any event in the city & submit it to JMC.

- **Around Me:**

This will have all the map layers (ATM, Banks, Primary Schools, Church, Fitness clubs.) available for the citizen to select & set the aerial distance (within which they wish to see the utilities)

- **Announcements:**

Any announcements made by JMC for citizens is to be displayed here

- **Add information:**

Provision for citizens to add image, data on the portal & submit it to JMC

- **Grievance & Feedback:**

Citizen can submit their grievances & feedbacks through this module

- **Help:**
Help regarding functionalities of all the features of the portal is to be found here
- **Language:**
A button for toggling the languages English and Gujarati
- **Heritage Sites:**
Heritage site' based tab (Representation on map with photo & Details) which helps citizens to look for heritage sites like Museums, Forts, Lakes
- **Print, Book Mark, Measure:**
Facility to Print, Bookmark & perform Measurement (Find Distance, Area & Latitude/Longitude)
- This portal as mentioned earlier, will host the repository of all data outputs created. The user will select multiple images which will be viewed in their respective Image viewers with the same zoom levels. These viewers will be geographically linked so that changes for the same location in different periods can be checked.
- Along with standard map functions such as zoom in, zoom out, pan, draw, print map, this web application will have Swipe, Time Series Analysis / Time Slider tools which will help the JMC administrators to easy detect the changes. Layer on-off should be provided too.
- The portal should use 2D interface for presenting 2D data and 3D interface for presenting any future 3D data.
- The bidder shall also provide Analysis & functionalities which will be able to process selected 2 datasets and produce a change detection locations and map.
- The bidder will also have to develop mobile based applications for Geo-enabled survey and other field-based verification as well as validation of the locations identified from change detection.
- The applications should develop in such a way that changes (Locations) identified on web portal can be directly assign to specific mobile application user to verify/collection of data from field.
- The dashboard should let administrators know, percentage of ward level statistics of changes identified

- Statistics on changed features and their timelines should be mapped properly so that it can be viewed and analyzed using this dashboard.
- This dashboard should also provide a summary of actions taken by the administrators and the subsequent visualizations should be configured.

4.3.7.3 Solution Design

The Bidder would prepare and submit the Software Requirement Specification document (SRS), Design Document, Software Design Document (SDD) for review and sign-off. Once the Design Document and SDD is signed-off, the bidder is expected to provide clickable and interactive wireframes, sample reports, sample dash boards. for approval before proceeding to final development. The Bidder would then carry out the further activities of software development / customization and UAT.

- The Bidder must ensure that the solution technology components adhere to flexibility, usability, availability, manageability, security and integration standards. Design should make use of common government components wherever possible. Design should comply with minimum functional requirements. Design should be flexible to adapt to changes required as per directives from JMC
- The Bidder shall build a complete audit trail of all transactions (i.e. add, update and delete) using transaction log reports, so that errors in data, intentional or otherwise, can be traced and reversed. The most appropriate level of security commensurate with the value to that function for which it is deployed must be chosen. Access Controls must be provided to ensure that the datasets are not tampered with or modified by the system operators. Implement data security to allow for changes in technology and business needs. Based on the requirements analysis conducted above, the Bidder must develop a comprehensive system.
- Design, development, testing and implementation of the Solution Design: The Bidder shall be entirely responsible for the architecture of the system implementation to satisfy all features, functions, performance, and security as described in this document including sizing of the required hardware and software. Solution description provided in this document is for reference only.

- The Bidder should ensure possible and required improvements. The Bidder shall consider users' inputs when they are finalizing all design components including user interfaces, mode of data entry, storage and retrieval, outputs reports, queries and the application design. All changes being done will be mutually discussed with concerned officials.
- System should have capability to integrate with external IT Systems of JMC.

4.3.7.4 Testing

The selected System Integrator shall undertake the following activities:

- Outline the methodology that will be used for testing the system
- Define the various levels or types of testing that will be performed for system
- Provide necessary checklist/documentation that will be required for testing the system
- Describe any techniques, test cases/ scenarios / scripts that will be used for testing the system
- Describe how the testing methodology will conform to requirements of each of the functionalities
- Indicate/demonstrate to the department that all applications installed in the system have been tested
- The selected System Integrator shall provide a workflow for on-off on test deliverables that will be mutually agreed upon by both the parties

In view of the above, different types of testing followed generally as a part of typical Software Development Life Cycle have been mentioned below to understand the exact testing requirements for this project:

- Unit testing - Testing of individual software components or modules. Typically done by the programmer and not by testers, as it requires detailed knowledge of the internal program design and code.
- Integration testing - Integration testing shall be amongst the modules to verify the functionality of data sharing within the application. This shall include integration

testing with external applications to verify the combined functionality after integration.

- Regression testing – After implementing a bug fix or new change request, the system shall be tested again by the to check whether any new bug has been introduced or not. Here the application shall be tested as a whole after each update or change in the software. The cost of any automation tools used in this testing shall be borne by the System Integrator.
- Acceptance testing - Normally this type of testing is done to verify if system meets the customer specified requirements. The department will conduct this testing based on the documents submitted by selected agency to determine whether to accept application.
- Usability testing – This is for the User-friendliness check. In this, following areas shall be tested: Application flow, for a new user understand the application early, proper help documented whenever a user is stuck at any point, minimum keyboard entry and maximum use of drop-downs/auto-prompt feature. Basically, system navigation shall be checked in this testing.
- Security testing – This test shall be conducted to check if the system can be penetrated by any hacker. The selected shall conduct security testing to check how well the system protects against unauthorized internal or external access and to check if system database is safe from external attacks.
- Performance testing – This term often used interchangeably with ‘stress’ and ‘load’ testing to check whether system meets performance requirements under different levels of loads and to check the maximum load at which the system performance degrades. For this testing, selected are expected to use different performance and load tools.
- Concurrency testing - Concurrency testing is also known as multi-user testing, performed to identify the defects in an application when multiple user’s login into the application. It helps in identifying and measuring the problems in system parameters such as response time, output, locks/dead locks or any other issues associated with concurrency. They shall carry out proper testing as required to meet the stated performance criteria.

4.3.7.5 Comprehensive audit including security Certification, from CERT-IN empaneled agency for the applications developed as per the scope of work

The SI shall conduct comprehensive audit including security Certification, from CERT-IN empaneled agency for the applications developed as per the scope of work. The comprehensive audit, VAPT should be done before Go-Live.

4.3.7.6 UAT, Training & Go-Live

➤ User Acceptance Testing

- The department shall form different user groups for the purpose of UAT. These user groups would test the application and the results of all the above-mentioned tests submitted by the selected System Integrator will be considered to ascertain the functionality of the application. Once the defined acceptance criteria are completely met, selected System Integrator shall take a formal sign off from the group for acceptance of each module. Based on the sign off and user feedback, department would issue UAT certification to the selected
- System Integrator for that module / sub-module. The selected System Integrator shall submit the Test Strategy, Test Plan and test Metrics to JMC.
- The Test Strategy document (prepared once for the project) shall include –
- Objectives of testing and the activities that will serve to meet these objectives
- ✓ Contents of the tests (which characteristics are to be tested and which are not)
- ✓ Defining priority levels and severity levels
- ✓ Associated documents (such as the project plan, overall quality assurance strategy, other project documents) being referenced
- ✓ Associated documents (such as the project plan, overall quality assurance strategy, other project documents) being referenced
- ✓ Resource staffing and allocation (hardware/software/personnel) for testing
- ✓ Test stages and cycles
- ✓ Selection of suitable testing methods
- ✓ Test management (defect reporting, defect management & trialing)

- ✓ Release environments and test tools used for testing
- ✓ The Test Plan document (prepared for every change/enhancement/patch) shall include
- ✓ Scope of testing within the context of the project
- ✓ Test systems and objects to be tested with priority levels
- ✓ Impact analysis in the system landscape – This shall include the identification of risks, either positive or negative, analyze the risk, the probability of the risk occurrence and its impact and the plan to control or mitigate the risk
- ✓ Criteria for interrupting and terminating testing, as well as acceptance criteria for each test stage i.e. Entry / Exit Criteria

➤ **Training& Capacity Building**

The selected System Integrator shall provide training to the application users to efficiently use the system as per requirement during the contract period.

After each training, training completion certificate duly signed by the trainees, trainers shall be maintained by the selected System Integrator.

➤ **Go-Live**

The Go-Live will be provided by JMC team to the SI on successful completion of development, installation, testing, bug-fixing, comprehensive audit, UAT and training.

➤ **Source Code Rights**

The rights of the source code of the customized version of the product and/or the bespoke application shall lie with JMC. In case any subsystem is completely developed, the rights of the source code shall also be with JMC. The source code needs to be transferred within three months of successful Operational Acceptance. All the costs associated with the transfer of source code shall be borne by the System Integrator. This shall also include customs or any other applicable taxes to be paid for import/export of software. The following protocol shall be followed for the transfer of source code to JMC

- The System Integrator shall submit detailed documentation for each module of the application to the department. This includes all the modules deployed. The lists of the documents to be submitted are as follows:
 - ✓ Functional Specification Document
 - ✓ Technical Specification Document
 - ✓ Software architecture Documentation
 - ✓ User Manuals
 - ✓ Training Manuals
 - ✓ Test Plan, Test Cases and Results
 - ✓ Release Notes
 - ✓ User Guide to install, configure and use the module

All the documents shall be updated as per the last release of that module. The documents shall be reviewed by the department or agency appointed by the department. System Integrator shall ensure that any disparity / lacunae found in the documents are rectified and revised documents are submitted for further review. The transfer of documentation to be considered as complete after the department issues the completion certificate for the task.

- Transfer of the source code files, supporting libraries, database scripts, libraries and metadata dictionary, procedures and supporting software components. Source Code to be exact replica of the department application live on the production server.
- Documentation of step-by-step procedure for recompilation of the application shall be submitted by System Integrator. The documentation shall enable JMC (or any third party appointed by JMC) to install, configure and recompile the application.
- While submitting the Source Code files, System Integrator shall submit the declaration that the Source code is of the same version which is on Production Environment and used for live operations. SI shall provide the environment to recompile the source code and provide access to the application to confirm

that the Source Code is of the latest version and is same as that on the Production Environment.

- System Integrator shall conduct the necessary Knowledge Transfer sessions to the technical staff provided by the JMC. The success criterion of training will be that IT team provided by JMC is able to recompile successfully the entire application on the test server independently.
- The transfer of source code shall be an on-going exercise. As and when, a new version of the application is deployed in production; the source code of the changed modules shall be transferred as per the above protocol to the department. At the end of contract period or at the end of the complete development and deployment of all the change requests provided by the department within the contract period or whichever is later, the entire source code shall be transferred in the same way.
- The SI should conduct (a) Software architecture and code review and (b) Security Audit of the application, and necessary compliance before handing over the source code during exit management. Timelines for this compliance shall be jointly decided between the department and System Integrator.

4.3.8 Hosting of all deliverables, raw datasets & Geo-spatial Application

The SI is expected to host all the deliverables, raw data, and applications on the JMC provided storage. The supply of cloud is not part of a SI's Scope. The Change Detection GIS application should be hosted on JMC server during implementation and AMC period.

4.3.9 Supply and Installation of Hardware

The bidders shall Supply, install & configure following software:

Sr. No	Hardware	Quantity
1	High End Desktop Workstations	05

4.3.9.1 Specifications of Hardware

The SI should install & commission five High-end workstations for viewing, editing 2D/3D data base and all the deliverables & raw data which is part of the scope of work this project. Below are the specifications of high-end workstation.

#	Parameter	Minimum Specifications
1.	Form Factor	All-in-One (AIO)
2.	Processor	3 GHz (8 core, 12 MB Cache or better) 9th Gen. Intel Core i7 or latest generation
3.	Memory	32 GB DDR4 RAM @ 2666 MHz. with additional DIMM Slot expandable to 128 GB
4.	Chipset	Compatible Chipset
5.	Graphics	Dedicated Graphics card with RAM 4 GB or more
6.	Hard Disk	2 TB or higher
7.	Audio	High-Definition Integrated Audio with combo headphone, Mic jack (Audio In & Out)
8.	Network port	10/100/1000 Mbps on-board integrated RJ-45 Ethernet Port
9.	Wireless Connectivity	Wireless LAN – 802.11 a/b/g/n/ac & Bluetooth
10.	USB Ports	Minimum 5 USB ports
11.	Display Port	1 Display, HDMI Port, 1 VGA
12.	Power supply	120 Watt with required cables and other accessories
13.	Keyboard	USB 104 Keys Keyboard of same make as PC
14.	Mouse	Optical mouse with USB interface of same make as PC
15.	Display	23" or above non-touch Full HD (1920 x 1080) resolution, integrated speakers (2 Numbers) with dedicated height adjustable stand and other accessories
16.	Web Camera	Inbuilt HD web camera with mic
17.	Operation System and Support	Pre-loaded Windows 11 (or latest) Professional 64 bit, licensed copy, Recovery media/partition

18.	Certification for Desktop	Energy Star 7 / BEE star, EPEAT, UL/FCC, CE, RoHS, Windows 11 certified
19.	Other pre-loaded software (COTS/free)	Latest version of MS-Office, Latest version of Adobe Acrobat Reader

4.3.10 Setting up GIS Centre of Excellence

- Maintenance of spatial and non-spatial database is critical since the quality of decisions made from a GIS depends largely on the quality of the GIS database. Hence, maintaining data quality and routinely updating the system is imperative. Therefore, the successful bidder shall assist the Authority in setting up a GIS Centre of Excellence for supervising and managing the GIS data during the implementation and operation & maintenance of the GIS.
- To update and manage the GIS Data & Application, Successful Bidders shall deploy two GIS Analyst with three-year experience for period of three year after GO-LIVE.

4.3.11 Capacity Building and Training

- The Successful Bidder shall provide training to departmental users to efficiently use the system. The staff thus trained would subsequently train the other staff as and when required.
- The Successful Bidder shall provide training as per the proposed training plan schedule to be shared as part of Approach and Methodology section in technical bid.
- The Successful Bidder shall provide training to the selected officials of Authority as decided by the authorized official. The training batch size should not be more than 25 officials.
- A detailed training schedule, including the dates, areas to be covered, time and the training literature (to be supplied to Authority) at various stages of the training cycle and feedback for effectiveness shall be agreed to by both parties (Authority and the Successful Bidder) during the performance of the Contract.
- Training to be imparted to users:

- ✓ **Functional Training:** This training would focus on the usage of application software so that the users are aware of all the operations of the application systems, ensuring a smooth run of Citizen Services or Departmental Operations. It would be covered for each of the functional module.
- ✓ **Administrative Training:** This training would focus on the administration of Application Software and Server Infrastructure and would be imparted to the relevant staff of Authority

4.3.12 Operation & Maintenance

Once the systems have been commissioned, the Successful Bidder shall provide three (3) years support for GIS data management, application management and future enhancements. The Bidder shall propose the teams for this along with their roles, job descriptions and profiles of key individuals as specified. The team must be operating from as required at site and bidders' office. The team shall perform but not limited to the following activities:

- Minor Enhancement / modifications with respect to new / enhanced / enriched functionality
- Ensure the desired functioning of the Interface / integration
- System installation and testing whenever required
- Provide technical support on system parameters and requirement of Authority
- Manage Enterprise GIS Data Edits
- Provide handholding support and training services as part of the post implementation services, on a scheduled basis as well as on a need basis.
- Process collected Data, undertake necessary Quality Assurance (QA) and Quality Checking (QC) activities, and remove errors
- Prepare updated comprehensive database including spatial and associated attribute Data and based on geographic coverage, richness, and accuracy on daily basis.
- The Successful bidder shall provide the below reports during the Operation & Maintenance phase:
 - o Monthly / Quarterly / Annual Performance Report of below:

- ✓ Detailed Work Report
- ✓ Downtime Report
- ✓ MIS Reports
- ✓ Resources' Work & Progress report

4.3.13 Project Timelines

Following is an Indicative Project Timeline:

Sr.No	Activity/Deliverable	Completion Timeline (Weeks)
1	Award of Contract & Team Mobilization	T
2	System Requirement Specification (SRS)	T+4
3	Functional Requirement Specification (FRS)	T+4
4	Procurement of Archival/Fresh satellite imagery	T+8
5	DGPS Survey for Ground Control Points (GCP)	T+8
6	Supply of Desktop Workstations (05 No's)	T+4
7	Supply of COTS based GIS and image processing Desktop software (05 No's)	T+4
8	Generation of Geo-referenced Ortho Mosaic Image	T+8
9	Delivery of 2D Base Map	T+16
10	Delivery of Change Detection Maps	T+16
11	Delivery of Underground utilities network	T+20
12	Supply of COTS based GIS and Image processing Enterprise Server Suite (16 core)(1 no's)	T+10
13	Development of Web GIS Application/Portal	T+12
14	Development of Mobile Application	T+12

15	Integration with other Systems of JMC	T+16
16	UAT & Go-Live	T+20
17	Training & Capacity Building	T+24
18	Setting up of GIS Cell and Operation & Maintenance Support	Three Years from Go-Live

4.3.14 Final Deliverables:

Below mentioned are the deliverables that needs to be delivered by the successful bidder.

Sl.no	Deliverable	Format
1	System Requirement Specification (SRS)	PDF
2	Functional Requirement Specification(FRS)	PDF
3	Geo Database Model	GDB
4	DGPS Ground Control Points	.SHP
5	Ortho Mosaic image	Geo-tiff
6	Ortho Mosaic Tiles(1x1 Sq km)	Geo-tiff
7	2D Base Map with underground utility network of pilot area	OGC compliant format
8	Change Detection Maps	OGC compliant format
9	Web GIS Application	
10	Mobile Application (POI & Property Survey)	.APK
11	Training Material	.pdf

4.3.15 Proposed Man Power:

Below Mentioned are the Manpower details that need to be required during the execution of the project.

#	Proposed Role	Qualification	Experience
1	Project Manager	M.Sc/M.Tech	5+Years of GIS Projects
2	GIS Analyst	B.Tech	3+Years
3	GIS Database Administrator	MCA/B.tech	3+Years
4	GIS Developer	B.Tech/MCA/M.SC	3+Years
5	Remote Sensing/Image Processing Expert	B.Tech/M.Tech	3+Years

5 Annexure I: Instructions for Pre-Qualification Bid

5.1 Pre – Qualification Cover Letter

Date: dd/mm/yyyy

To

<Insert complete postal address>

Sub : Selection of Agency to provide Technical Manpower services in GIS and Remote Sensing in JMC

Ref : Bid No: <No> Dated <DD/MM/YYYY>

Dear Sir,

Having examined the bid, the receipt of which is hereby duly acknowledged, we, the undersigned, offer to provide the professional services as required and outlined in the

bid for the *Design, Development, Implementation, and Maintenance of Spatially enabled Web Portal using latest Technology for Jamnagar Municipal Corporation.*

We attach hereto our responses to pre-qualification requirements and technical and commercial bids as required by the bid. We confirm that the information contained in these responses or any part thereof, including the exhibits, and other documents and instruments delivered or to be delivered to Jamnagar Municipal Corporation, are true, accurate, verifiable, and complete. This response includes all information necessary to ensure that the statements therein do not in whole or in part mislead the Corporation in its short-listing process.

We fully understand and agree to comply that on verification, if any of the information provided here is found to be misleading the selection process, we are liable to be dismissed from the selection process or termination of the contract during the project, if selected to do so.

We agree for unconditional acceptance of all the terms and conditions set out in the bid document and also agree to abide by this bid response for the bid validity period as mentioned in the RFP. We hereby declare that in case the contract is awarded to us, we shall submit the contract performance guarantee bond in the form prescribed in the bid.

We agree that you are not bound to accept any bid response you may receive. We also agree that you reserve the right in absolute sense to reject all or any of the products/ services specified in the bid response.

It is hereby confirmed that I/We are entitled to act on behalf of our company/ corporation/ firm/ organization and empowered to sign this document as well as such other documents, which may be required in this connection.

Signature of Authorized Signatory (with official seal)

Name :

Designation :

Address :

Telephone :

Email address :

5.2 Format to share SI's Particulars

Sr. No.	Description	Details (to be filled by the responder to the Bid)
1.	Name of the company	
2.	Official address	
3.	Phone No.	
4.	Corporate Headquarters Address	
5.	Phone No.	
6.	Web Site Address	
7.	Details of Company's Registration (Please enclose copy of the company registration document)	
8.	Name of Registration Authority	
9.	Registration Number and Year of Registration	
10.	ISO/CMMi certifications and its validity	
11.	CST/LST/VAT registration No.	
12.	Service Tax /GST Registration No.	
13.	Permanent Account Number (PAN)	

14.	Company's Revenue for last 3 years (Year wise)	
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Please submit the relevant proofs for all the details mentioned above along with your bid response.

Contact Details of officials for future correspondence regarding the bid process:

Details	Authorized Signatory	Secondary Contact
Name		
Title		
Company Address		
Phone		
Mobile		
Email		

5.3 Format for Declaration by the SI for not being Blacklisted / Debarred

(To be submitted on Stamp Paper of ₹ 500)

DECLARATION CUM-INDEMNITY BOND

Date: dd/mm/yyyy

I, _____ of _____, do hereby declared and undertake as under.

1) I declared that I have submitted certificates as required to Executive Engineer (Monitoring) at the time of registration of my firm / company _____ and there is no change in the contents of the certificates that are submitted at the time of registration.

2) I declared that I _____ in capacity as Manager / Director / Partners / Proprietors of _____ has not been charged with any

prohibitory and /or penal action such as demotion, suspension, black listing / de-registration or any other action under the law by any Government and / or Semi Government and/ or Government Undertaking.

3) I declared that, I have perused and examined the tender document including addendum, condition of contract, specification, drawings, bill of quantity etc. forming part of tender and accordingly, I submit my offer to execute the work as per tender documents at the rates quoted by me in capacity as _____ of _____.

4) I further declared that if, I am allotted the work and I failed to carry out the allotted work in accordance with the terms and conditions and within the time prescribed and specified, JMC is entitled to carry out the work allotted to me by any other means at my risk and cost, at any stage of the contract.

5) I also declared that I will not claim any charge / damages / compensation for non-availability of site for the contract work at any time.

6) I Indemnify Municipal Commissioner and the other officers of JMC or their agents for any Damages, Loss, or Injury, any legal suit, proceeding or legal action whatsoever that may be caused at any time by me or any other staff of _____ company, for the work undertaken and all such damage, damages, injury or loss, legal suit, legal action, I shall be solely responsible in individual as well as official capacity and such loss, damages, injury shall be made good and/ or as the case may be shall be paid immediately by me / Company to the satisfaction of the JMC.

Dated _____ day of _____, 20__

Identified by me

Before me

Advocate

6 Annexure II: Instructions and Technical Bid Document Formats

6.1 General Instructions for the Technical Bid

Sis have to submit a very structured and organized technical bid, which will be analysed by the Technical Evaluation Committee for different compliances with

regards to the requirements of the project. Since the cut-off marks for Technical bid Score is 70, the quality and completeness of the information submitted by the SI will matter a lot.

SI is expected to divide its Bid in following sections / documents:

3. SI's Competence to execute the project

This document should bring about the capability of the firm to execute this project. Some of the required documents are as follows:

- a) Financial Capability of the SI in required formats and supporting documents
- b) Experience of SI of implementing similar/same applications or in supply of similar/same hardware components if applicable
- c) Experience of SI in India in required formats and supporting documents
- d) Quality of GIS + IT and domain experts available with the firm

B) Proposed Team to be deployed at JMC for maintenance

As specified in the Technical Bid Evaluation Framework, the department would like to give importance to the desired / right people proposed for deployment. SI may propose different people for different skill-sets required and different responsibilities (during Maintenance Support and Project Implementation). Following documentation is expected in this section:

- a) Overall Project Team (for both Project Implementation and Support phases)
- b) Summary Table giving Qualification, Experiences, Certifications, Relevance
- c) Detailed CVs in the format attached
- d) Escalation Chart for the entire Project Duration

C) Technical Solution Proposed for the Project Broad areas to be covered in the Technical Solution documentation are given below:

- a) Clearly articulate the Strategy and Approach and Methodology for the work to be executed by the resources in line with the scope of services expected.

- b) Approach and Methodology for Management of SLA Requirements specified in the bid. SI is required to clearly articulate how the SLA requirements would be adhered to.
- c) Detailed Project Plan with timelines, resource allocation, milestones etc.
- d) Clearly mention the risk and mitigation plan.

6.2 Format to share Project Details

Name of the Project	
General Information	
Client for which the project was executed	
Name of the client contact person(s)	
Designation of client contact person(s)	
Contact details of the client contact person(s)	
Project Details	
Description of the project	
Scope of work of the SI	
Deliverables of the SI	
Technologies used	
Current Status of the project	
Other Details	
Total cost of the project	
Total cost of the services provided by the SI	

Duration of the project (number of months, start date, completion date, current status)	
Other Relevant Information	
Mandatory Supporting Documents: 1. Work order / Purchase order / Contract for the project 2. Client Certificate giving present status of the project and view of the quality of services by the SI	

Note: The SI is required to use above format for all the projects referenced by the SI for the pre-qualification criteria and technical bid evaluation.

6.3 Details of Manpower Resources Proposed

#	Name of the Resource	Proposed Role	Highest Qualification	Total Experience (in years)	Total Relevant Experience for the proposed position (in years)

CV of the Manpower proposed to be submitted in the following format:

1	Name of the Staff	
2	Current Designation in the Organization	
3	Proposed Role in	

	the Project					
4	Proposed Responsibilities in the Project					
5	Date of Birth					
6	Education	Degree / Diploma, College, University, Year of Passing				
7	Summary of Key Training and Certifications					
8	Language Proficiency					
		Language	Re adi ng	W riti ng	Sp eak ing	
9	Employment Record (For the total relevant experience)					
		From / To:				
		Employer:				
		Position Held:				
		From / To:				
		Employer:				
		Position Held:				
From / To:						

		Employer:		
		Position Held:		
10	Total No. Of Years of Work Experience			
11	Total No. Of Years of Experience for the Role proposed			
12	Highlights of relevant assignments handled and significant accomplishments (Use following format for each project)	Name of assignment or project:		
		Year:		
		Location:		
		Client:		
		Main project features:		
		Positions held:		
		Activities performed:		

Important Note:

SIs need to submit the soft copies of following along with the complete resume:

1. Certificates of educational qualifications
2. Authorized person submitting the bid needs to sign the above as in the case of other bid submission related documents.

7 Annexure III: Commercial Bid Format

The Bidder must quote the rate in the BoQ Spreadsheet available online with this bid.

Details to be filled up for price bid are as below. The fees shall be inclusive of Goods & Service Tax (GST) as applicable under the relevant Laws of India.

7.1 Implementation Cost

Sr.No	Activity/Deliverable	Unit	Quantity	Unit Cost (INR)	Total Cost (INR)
C1	Procurement of Archival/Fresh(2025) satellite imagery(Stereo)	Sq Km	150		
C2	DGPS Survey for Ground Control Points(GCP)	No's	150		
C3 (a)	Supply of Desktop COTS GIS software and Workstations	No s	5		
C3 (b)	Supply and Installation of COTS based Enterprise GIS and Image processing software (16 Core)	No s	1		
C4	Delivery of Geo-referenced Ortho Mosaic Image	Sq Km	150		
C5	Delivery of 2D Base Map	Sq Km	150		
C6	Change Detection maps	Sq Km	150		
C7	Underground Utility Survey	Km	10		
C8	Development of Web GIS Application/Portal 2D and 3D Compatible on Open-Source platform	No	1		
C9	Development of Mobile Application(Android &IOS)	No	1		

C10	Integration with existing Systems of JMC	No	1		
Total					
Total Amount in Words					

7.2 Operation & Annual Maintenance Cost

Operation & Maintenance period for 3Years.

Sr.No	Activity/Deliverable	Unit	Quantity	Unit Cost (INR)	Total Cost (INR)
O1	Onsite Support GIS Cell Resources (5 No's)	Months	36		
O2	Procurement of Fresh satellite imagery(Stereo Pair) on Yearly Interval from 2026 to 2028	Sq Km	450		
O3	Change Detection reports	Sq Km	450		
O4	AMC of Hardware	Months	36		
O5	AMC of Bug fixing, issue resolution & enhancements of the Enterprise Geospatial applications	Months	36		
Total					

8 Annexure IV: Draft Contract Agreement

On a Stamp Paper of Rs.500/- for contract value up to Rs.10,00,000/- and 0.1% of the amount above Rs.10 lakh subject to maximum Rs.25 Lakhs)

CONTRACT AGREEMENT

Bid No :- _____

Sanction No. & Date: _____

Contract Cost : _____

This Contract Agreement (hereinafter referred to as “CA”) is made and entered into at Jamnagar this

_____ day of _____ Two Thousand Nineteen

BETWEEN

The Jamnagar Municipal Corporation, a body corporate having perpetual succession and common seal constituted by the Jamnagar Municipal Corporation Act, 1888, hereinafter referred to as “The JMC”

REPRESENTED BY

Shri. _____, Director (Information Technology), having its office at _____, hereinafter referred to as “Director(IT)” (which expression shall unless repugnant to the context or meaning thereof be deemed to mean and include the successor or successors for the time being holding the office of the Director (IT) of the First Part;

AND

M/s. _____, a company incorporated under the provisions of the Companies Act, 1956 and having its registered office at _____ through Shri. _____, (Authorized Signatory); hereinafter referred to as “the Consultant” (which expression shall unless repugnant to the context or meaning thereof be deemed to mean and include its successor or successors and assigns) of the Second Part;

WHEREAS the Municipal Commissioner of JMC has inter alia deputed under Section 56 of the Jamnagar Municipal Corporation Act, 1888 his powers, functions and duties to the Director(IT) / AMC(ES) for signing and executing this Contract Agreement on behalf of JMC.

AND WHEREAS M/s. _____, by its Resolution under No. _____ dated _____ have authorized Shri _____ (Authorized Signatory), to sign the present Contract Agreement, on behalf of the said Company.

AND WHEREAS the JMC has invited Tender for Selection of Agency to Development, Implementation, Maintenance of Change Detection using Geospatial Technology for JMC; hereinafter referred to as “the contract work”. The System Integrator has submitted the Tender for the said contract work.

AND WHEREAS the Standing Committee vide its Resolution under no. _____ dtd. _____ has accorded sanction to _____, as Implementation Agency for Development, Implementation, Maintenance of Change Detection using Geospatial Technology for JMC, for a period of _____ years, at total cost of Rs. _____/- (including 18% GST) on the terms and conditions mentioned therein.

AND WHEREAS the Implementation Agency has agreed to comply with the terms and conditions hereinafter appearing and are desirous of recording the same subject to compliance of which the contract as aforesaid has been agreed to be granted by the JMC in favor of the Implementation Agency.

NOW THIS CONTRACT AGREEMENT WITNESSETH AND IT IS HEREBY AGREED BY AND BETWEEN THE PARTIES HERETO AS FOLLOWS:

1. The parties hereto agree that the recitals enumerated herein above shall be deemed to form an integral and operative part of this Contract Agreement as if the same were specifically set out and incorporated herein.
2. The following documents are and shall be deemed to form part of this Contract Agreement and shall be read and construed to be part of this Contract Agreement as if they were incorporated in this Contract Agreement.

- (i) Corrigendum document published by JMC subsequent to the Bid for this contract work.
- (ii) Bid Document of JMC issued for this contract work under no. _____
- (iii) Standing Committee Resolution no. _____ dtd. _____
- (iv) Letter of Acceptance issued under no. _____ dtd. _____
- (v) Special Conditions of Contract
- (vi) General Conditions of Contract

8.1 Definitions, Interpretations and Other Terms

- ✓ “Bid” means the bid process conducted by JMC including all clarifications/addendums, explanations and amendments issued by the Corporation in respect thereof and the technical and commercial bids submitted by the successful SI, along with the subsequent clarifications and undertakings, if any.
- ✓ The “Contract” shall mean the tender and acceptance thereof and the formal Agreement if any, executed between the SI, Commissioner, and the Corporation together with the documents referred to therein including these conditions and appendices and any special conditions, the specifications, designs, drawings, price schedules, bills of quantities and schedule of rates. All these documents taken together shall be deemed to form one Contract and shall be complementary to one another.
- ✓ The “SI” shall mean a person or corporate body who has desired to submit Bid to carry out the Works, including routine maintenance till the tender process is concluded.
- ✓ “Arbitrator” means the person or persons appointed by agreement between JMC and the SI to make a decision on or to settle any dispute or difference between the JMC and the SI referred to him or her by the parties pursuant to General Conditions of Contract amended up to date.
- ✓ “Contract Cost” means the Contract Sum plus Price Variation. This cost shall be included in the letter of acceptance.
- ✓ “Excepted risks” are risks due to riots (otherwise than among SI’s employees) and civil commotion (in so far as both these are uninsurable), war (whether

declared or not), invasion, act of foreign enemies, hostilities, civil war, rebellion, revolution, insurrection, military or usurped power, any act of government, damage from aircraft, acts of god, such as earthquake, lightning and unprecedented floods and other causes over which the SI has no control and accepted as such by the Commissioner or causes solely due to use or occupation by the Municipal Corporation of the works in respect of which a certificate of completion has been issued or a cause solely due to faulty municipal design of work.

- ✓ The “Corporation” or the “Municipal Corporation” shall mean the Jamnagar Municipal Corporation
- ✓ The “Annexure” referred to in these conditions shall mean the relevant annexure appended to the tender papers issued by the Municipal Corporation.
- ✓ The “Works” shall mean the Permanent Works and the Temporary Works or either of them as appropriate to be executed in accordance with the contract or part(s) thereof, as the case may be and shall include all extra or additional, altered or substituted works as required for performance.
- ✓ “Temporary Works” shall mean all Temporary Works of every kind required in or about execution, completion or maintenance of the work also Temporary Works are works designed, constructed, installed, and removed by the Service Provider that are needed for construction or installation of the Works of the contract as found necessary as per suggestion of the JMC.
- ✓ “Permanent Works” means the permanent works to be executed and installation of machineries in accordance with the Contract at specified required site and location.
- ✓ “Drawings” means all the drawings, calculations and technical information of a like nature provided by the JMC to the Service Provider under the Contract and all drawings, calculations, samples, patterns, models, operation & maintenance manual and other technical information of like nature submitted by the Service Provider and approved by the JMC.
- ✓ “Approved” shall mean approved in writing including subsequent confirmation of previous verbal approval and “approval” shall mean approval in writing including as aforesaid.

- ✓ “Specification” means the specification referred to in the tender and any modification thereof or addition or deduction thereto as may from time to time be furnished or approved in writing by the JMC
- ✓ “Tender” means the SI’s priced offer to the JMC for the execution and completion of the Works and the remedying of any defects therein in accordance with the provision of the Contract, as accepted by the Letter of Acceptance.
- ✓ “Commencement Date” means the date upon which the SI receives the notice to commence, issued by the JMC.
- ✓ “Time for Completion” means the time for completing the execution of and passing the Tests on Completion of the Works or any Section or part thereof as stated in the Contract calculated from the Commencement Date.
- ✓ “Certificate of Taking-Over/ Take-over Certificate” shall mean the certificate issued by JMC after completion of Works in all respects.
- ✓ “Letter of Acceptance” means the letter of formal acceptance, signed by JMC, of the Letter of Bid, including any annexed memoranda comprising agreements between and signed by both Parties. If there is no such letter of acceptance, the expression “Letter of Acceptance” means the Contract Agreement and the date of issuing or receiving the Letter of Acceptance means the date of signing the Contract Agreement.
- ✓ “Cost” means all expenditure properly incurred or to be incurred whether on or off the site including overheads and other charges properly allocable thereto but does not include any allowance for profit.
- ✓ “Confidential Information” means all information including JMC Data (whether in written, oral, electronic or other format) which relates to the technical, financial, business affairs, customers, suppliers, products, developments, operations, processes, data, trade secrets, design rights, know-how and personnel of each Party and its affiliates which is disclosed to or otherwise learned by the other Party in the course of or in connection with this CA (including without limitation such information received during negotiations, location visits and meetings in connection with this CA).
- ✓ “Customers” mean all citizens and business organization and users who use the JMC services.

- ✓ “Deliverables” means all the activities related to the setting up and operations of the infrastructure, technical documents, Software Applications, customized Source Codes, as defined in the bid and subsequent Corrigendum (if any), based on which the technical bid and commercial bid was submitted by the SI and as required as per this CA.
- ✓ “Effective Date” means the date on which the Purchase Order or Letter of Intent is issued to System Integrator.
- ✓ “CA” means this Contract Agreement, together with the recitals and all schedules and the contents, requirements, specifications and standards of the bid (as may be amended,
- ✓ supplemented or modified in accordance with the provisions hereof) and the bid. In the event of a conflict between this CA and the Schedules, the terms of the CA shall prevail; with overriding effect.
- ✓ “Performance Security” means the irrevocable and unconditional Bank Guarantee provided by the System Integrator from by any of the approved banks as specified in Annexure V of this RFP document in favour of “Jamnagar Municipal Corporation” for an amount equivalent to 10% of the total contract value i.e.
☐(Rupees.....only) for due and faithful compliance of the contract work during the contract period.
- ✓ “Proprietary Information” means processes, methodologies, and technical and business information, including drawings, designs, formulae, flow charts, data and computer programs already owned/licensed by either Party or granted by third parties to a Party hereto prior/ subsequent to the execution of this CA.
- ✓ “Required Consents” means the written consents, clearances and licenses, rights and other authorizations as may be required to be obtained by the System Integrator, for all tasks/activities/software/hardware and communication technology for this project; from all the concerned departments/agency. as the case may be.
- ✓ “Bid Document” means the Request For Proposal released vide Bid document number specified under section 4.1, and include all clarifications/ addendums, explanations and amendments issued by the JMC in respect thereof;
- ✓ “Service Level(s)” means the performance standards, which will apply, to the services delivered by selected System Integrator.

- ✓ “Service Level Requirement(s)” means the timelines and the quality levels to be adhered to by the System Integrator for delivering various services under the contract.
- ✓ “Services” means the content and services delivered and to be delivered to the customers or the offices of JMC by the System Integrator, and includes but not limited to the services specified in the bid document or as may be specified and incorporated in the subsequent Agreement(s) under Contract Agreement.
- ✓ “Users” means the departmental staffs or any other JMC officials having access to application including its System Integrator, technology vendors, corporations and agency and their employees, as the context admits or requires.
- ✓ “Day” means Calendar Day.
- ✓ “Month” means Calendar month of the Gregorian calendar.
- ✓ “GCC” means General Conditions of Contract.
- ✓ “Foreign Currency” means currency of a country other than that in which the works are to be located, approved by Govt. of INDSI/Reserve Bank of INDSI.
- ✓ “Writing” means any hand written, type-written, or printed communication including telex and facsimile transmission, electronic and digital media.
- ✓ “Country” means the country in which the Site is located.
- ✓ “Variation” means a change to the: -
 - Specification and /or Drawings (if any) which is instructed by the JMC
 - Scope in the Contract which is instructed by the JMC
 - Price in the Contract which is instructed by the JMC
- ✓ “Specification” shall mean the specification referred to in the tender and any modification thereof or addition or deduction thereto as may from time to time be furnished or approved in writing by the JMC.
- ✓ “The Start Date/Commencement Date” is given in the Contract Data. It is the date when the Service Provider shall commence execution of the Works. It does not necessarily coincide with any of the Site Possession Dates.

8.2 Interpretations

- a. References to any statute or statutory provision include a reference to that statute or statutory provision as from time to time amended, extended, re-enacted or consolidated and to all statutory instruments made pursuant to it.
- b. Words denoting the singular shall include the plural and vice-versa and words denoting persons shall include firms and corporations and vice versa.
- c. Unless otherwise expressly stated, the words “herein”, “hereof”, “hereunder” and similar words refer to this CA as a whole and not to any particular Article, Schedule. The term Articles, refers to Articles of this CA. The words “include” and “including” shall not be construed as terms of limitation. The words “day” and “month” mean “calendar day” and “calendar month” unless otherwise stated. The words “writing” and “written” mean “in documented form”, whether electronic or hard copy, unless otherwise stated.
- d. The headings and use of bold type in this CA are for convenience only and shall not affect the interpretation of any provision of this CA.
- e. The Schedules to this CA form an integral part of this CA and will be in full force and effect as though they were expressly set out in the body of this CA.
- f. Reference at any time to any agreement, deed, instrument, license or document of any description shall be construed as reference to such agreement, deed, instrument, license or other document as the same may be amended, varied, supplemented, modified or suspended at the time of such reference.
- g. References “roll out” includes, unless the context otherwise requires, design, development, implementation, engineering, procurement, delivery, transportation, installation, processing, fabrication, acceptance testing, certification, commissioning and other activities incidental to the construction or roll out, and “construct” or “roll out” shall be construed accordingly.
- h. Any word or expression used in this CA shall, unless defined or construed in this CA, bear its ordinary English language meaning.
- i. The damages payable by a Party to the other Party as set forth in this CA, whether on per diem basis or otherwise, are mutually agreed genuine pre-estimated loss and

liquidated damages likely to be suffered and incurred by the Party entitled to receive the same and are not by way of penalties.

j. This CA shall operate as a legally binding agreement specifying the master terms, which apply to the Parties under this agreement and to the provision of the services by the System Integrator,

k. The Corporation may nominate a technically competent agency/individual(s) for conducting acceptance testing and certification of the various requisite infrastructure to ensure a smooth, trouble free and efficient functioning of the Scheme or carry out these tasks itself.

l. The agency/individual nominated by the Corporation can engage professional organizations for conducting specific tests on the software, hardware, networking, security and all other aspects.

m. The agency/individual will establish appropriate processes for notifying the System Integrator of any deviations from the norms, standards or guidelines at the earliest instance after taking cognizance of the same to enable the System Integrator to take corrective action.

n. Such an involvement of and guidance by the agency/person will not, however, absolve the System Integrator of the fundamental responsibility of designing, installing, testing and commissioning the application and the infrastructure for efficient and effective delivery of services as contemplated under this bid.

o. The following documents forming this Contract Agreement are to be taken as mutually explanatory of one another. The following order shall govern the priority of documents constituting this Contract Agreement, in the event of a conflict between various documents, the documents shall have priority in the following order:

i. This Contract Agreement;

ii. Standing Committee Resolution No. ____ dtd. _____;

iii. Purchase Order or Letter of Acceptance issued under no. ____dtd. ____issued by JMC to the System Integrator.

iv. Clarification & Corrigendum Documents published by JMCsubsequent to the Bid Document for this work

v. Bid Document of JMC for this work

vi. System Integrator's "Technical Proposal" and "Commercial Proposal" submitted in response to the Bid

8.3 Term of the Contract Agreement

(a) The term of this CA shall be a period of 4.5 (four & half) years from the date of issue of Letter of Acceptance/ Purchase Order i.e. _____. This includes the time required for Selection of Agency to provide Technical Manpower services for GIS and Remote Sensing in JMC.

(b) In the event of implementation period getting extended beyond the stipulated time, for reasons not attributable to the System Integrator, JMC reserves the right to extend the term of the Agreement by corresponding period to allow validity of contract from the date of successful go live.

(c) JMC also reserves the right to extend the contract at its sole discretion, beyond the initial four & half (4.5) years contract period with approval of the competent authority, and not otherwise.

8.4 JMC's Decision

Except where otherwise specifically stated, the Director IT, JMC will decide contractual matters between JMC and the SI in the role representing the JMC. However, if JMC's representative is required under the rules and regulations and orders of the JMC to obtain prior approval of some other authorities for specific actions, he will so obtain the approval, before communicating his decision to the SI.

Except as expressly stated in the Contract, JMC's representative shall not have any authority to relieve the Contractor of any of his obligations under the contract.

8.5 Delegation

JMC's representative, with the approval of the JMC Authority, may delegate any of his duties and responsibilities to other person(s), except to the Adjudicator, after notifying the SI, and may cancel any delegation after notifying the SI.

8.6 Communication

All certificates, notices, or instructions to be given to the SI by JMC shall be sent on the address / Email Address or contact details given by the SI of Bid. The address and

contact details for communication with the JMC shall be as per the details given in Contract Data. Communications between parties that are referred to in the conditions shall be in writing. The Notice sent by electronic means shall be effective on confirmation of the transmission. The Notice sent by Registered post or Speed post shall be effective on delivery or at the expiry of the normal delivery period as undertaken by the postal service.

8.7 Personnel

The Service Provider's personnel shall appropriately be qualified, skilled and experienced in their respective trades or occupations. JMC shall have authority to remove, or cause to be removed, any person employed on the site or works, who carries out duties incompetently or negligently.

If JMC asks the Service Provider to remove a person who is a member of the Service Provider's staff or work force, stating the reasons, the Service Provider shall ensure that the person leaves the Site within seven days and has no further connection with the Works in the Contract.

8.8 JMC's and Service Provider's Risks

JMC carries the risks which this Contract states are JMC's risks, and the Service Provider carries the risks which this Contract states are Service Provider's risks

8.9 JMC's Risks

The JMC is responsible for the excepted risks which are:

in so far as they directly affect the execution of the Works, the risks of war, invasion, act of foreign enemies, rebellion, revolution, insurrection or military or usurped power, civil war, riot, commotion or disorder (unless restricted to the Service Provider's employees) and contamination from any nuclear fuel or nuclear waste or radioactive toxic explosive, or

b. a cause due solely to the design of the Works, other than the Service Provider's design.

8.10 Service Provider's Risks

All risks of loss of or damage to physical property and of personal injury and death which arise during and in consequence of the performance of the Contract other than the excepted risks, are the responsibility of the Service Provider.

8.11 Management Meetings

JMC may require the Service Provider to attend a management meeting. The business of a management meeting shall be to review the plans for progress of the Works.

JMC shall record the business of management meetings and provide copies of the record to those attending the meeting. The responsibility of the parties for actions to be taken shall be decided by JMC either at the management meeting or after the management meeting and stated in writing to all those who attended the meeting.

8.12 Tests

The Service Provider will have to perform the mandatory tests as prescribed in the specifications.

The Service Provider shall be solely responsible for:

- a. Carrying out the mandatory tests prescribed in the Specifications, and
- b. For the correctness of the test results, whether preformed

If JMC instructs the Service Provider to carry out a test not specified in the Specification / Quality Assurance Handbook to check whether any work has a Defect and the test shows that it does, the Service Provider shall pay for the test and any samples. If there is no defect, the test shall be a compensation event.

The cost of testing shall be borne by the Service Provider even if the result of the sample confirms or do not confirm to the relevant specifications.

All expenditure required to be incurred in testing shall be borne by the Service Provider himself.

8.13 Variations

JMC shall, having regard to the scope of the Works and the sanctioned estimated cost, have power to order, in writing, Variations within the scope of the Works he/she considers necessary or advisable during the progress of the Contract. Such Variations shall form part of the Contract and the Service Provider shall carry them out and include them in updated Programmes produced by the Service Provider. Oral orders of JMC for Variations, unless followed by written confirmation, shall not be taken into account.

8.14 Payment for Variations

If rates for Variation items are specified in the Bill of Quantities, the Service Provider shall carry out such work at the same rate.

8.15 Operations and Maintenance Manuals

If Drawings and /or operation and maintenance manuals are required, the Service Provider shall supply them by the dates stated in the Contract Data.

If the Service Provider does not supply the Drawings and /or manuals by the dates stated in the contract Data, or they do not receive the JMC's approval, JMC shall withhold the amount stated in the Contract Data from payments due to the Service Provider.

8.16 Compliance with Labour Regulations

During continuance of the Contract, the System Integrator shall abide at all times by all existing labour enactments and rules made there under, regulations, notifications and bye laws of the State or Central Government or local authority and any other labour law (including rules), regulations, bye laws that may be passed or notification that may be issued under any labour law in future either by the State or the Central Government or the local authority.

Furthermore, the System Integrator shall keep JMC indemnified in case any action is taken against the JMC by the competent authority on account of contravention of any of the provisions of any Act or rules made there under, regulations or notifications including amendments. If JMC is caused to pay or reimburse, such amounts as may be necessary to cause or observe, or for nonobservance of the provisions stipulated in the notifications/ bye laws/ Acts/ Rules/ regulations including amendments, if any, on the part of the System Integrator, JMC shall have the right to deduct any money due to the System Integrator including his amount of performance guarantee.

The JMC shall also have right to recover from the System Integrator any sum required or estimated to be required for making good the loss or damage suffered by JMC.

The System Integrator shall require his employees to obey all applicable laws, including those concerning safety at work. The entire responsibility of the employees of the System Integrator, shall be of the System Integrator alone and under no

circumstances the employees of the System Integrator shall be treated as the employees of the JMC at any point of time.

8.17 Official Secrecy

The Service Provider shall of all the persons employed in any works in connection with the contract that the India Official Secrets Act 1923 (XIX of 1923) applies to them and will continue to apply even after execution of the said works and they will not disclose any information regarding this contract to any third party. The Service Provider shall also bring into notice that, any information found to be leaked out or disclosed the concern person as well as the Service Provider will be liable for penal action; further the Corporation will be at liberty to terminate the contract without notice.

8.18 Subsequent Legislation

If on the day of submission of bids for the contract, there occur changes to any National or State statute, Ordinance, decree or other law or any regulation or By-laws or any local or other duly constituted authority or the introduction of any such National or State Statute, Ordinance, decree or by which causes additional or reduced cost to the Service Provider, such additional or reduced cost shall, after due consultation with the Service Provider, be determined by the concerned authority of JMC and shall be added to or deducted from the Contract Price with prior approval of competent authority and JMC shall notify the Service Provider accordingly. JMC reserve the right to take decision in respect of addition/reduction of cost in contract.

8.19 Patent, Rights & Royalties

The Service Provider shall save harmless and indemnify the Corporation from and against all claims and proceedings for or on account of infringement of any Patent rights, design trademark or name of other protected rights in respect of any constructional plant, machine work, or material used for or in connection with the Works or any of them and from and against all claims, proceedings, damages, costs, charges and expenses whatsoever in respect thereof or in relation thereto. Except where otherwise specified, the Service Provider shall pay all tonnage and other royalties, rent and other payments or compensation, if any, required for the works.

8.20 Payment, Taxes & Claims

- ✓ **The limit for unforeseen Risks**

Under no circumstances whatever the Service Provider shall be entitled to any compensation from JMC on any account unless the Service Provider shall have submitted a claim in writing to JMC within One (1) month of the case of such claim occurring.

✓ **No interest for delayed payments due to disputes, etc.**

It is agreed that the Jamnagar Municipal Corporation or its representative or Officer shall not be liable to pay any interest or damage with respect of any money or balance which may be in its or its representative's or officer's hands owing to any dispute or difference or claim or misunderstanding between the Municipal Corporation of Greater Bombay or its representative or Officer on the one hand and the Service Provider on the other, or with respect to any delay on the part of the Municipal Corporation of Greater Bombay or representative or Officers in making periodical or final payments or in any other respect whatever.

8.21 Proprietary data

All documents and other information supplied by the Authority or submitted by a Service Provider to JMC Authority shall remain or become the property of JMC Authority. Service Provider to treat all information as strictly confidential and shall not use it for any purpose other than for preparation and submission of their Application. JMC Authority will not return any Application or any information provided along therewith.

8.22 Taxes

a) G.S.T. and other state levies / cess which are not subsumed under GST will be applicable. The tenderer shall quote inclusive of all taxes. It is clearly understood that JMC will not bear any additional liability towards payment of any Taxes & Duties.

b) Wherever the services to be provided by the Tenderers falls under Reverse Charge Mechanism, the price quoted shall be exclusive of GST, but inclusive of Taxes/ Duties/ Cess other than GST, if any.

c) Rates accepted by JMC shall hold good till completion of work and no additional individual claim shall be admissible on account of fluctuations in market rates, increase in taxes/ any other levies/ tolls etc. except that payment/ recovery for overall market situation shall be made as per price variation.

8.23 Contract Execution

All required documents for execution of the contract shall be submitted within 30 days from the date of issue of letter of acceptance. If the documents are not submitted within the stipulated time a penalty of ₹ 5000/- per day will be applicable to the SI. All contract documents need to be duly affixed with stamp duty properly signed along with evidence/proof of payment of security/contract deposit/ within 30 days from the date of letter of acceptance received by him.

If the amount of the Contract Deposit to be paid above is not paid within 30 days from the date of issue of Letter of Acceptance, the Tender / Contractor already accepted shall be considered as cancelled and legal steps be taken against the SI for recovery of the amounts.

The amount of Performance Bank Guarantee (PBG) retained by the JMC shall be released after expiry of period up to which the contractor has agreed to maintain the work in good order is over.

In the event of the contractor failing or neglecting to complete the rectification work within the period up to which the contractor has agreed to maintain the work in good order, the amount of Performance Bank Guarantee (PBG) retained by JMC shall be adjusted towards the excess cost incurred by the Department on rectification work.

In case of pending execution of contract agreement, Payment of 10% of Total contract value will be withheld.

8.24 Contract may be rescinded and Performance Bank Guarantee (PBG) forfeited for bribing a public officer or if contractor becomes insolvent

If the contractor assigns or sublets his contracts or attempt so to do, or become insolvent or commence any proceeding to get himself adjudicated and insolvent or make any composition with his creditors, or attempt so to do or if bribe, gratuity, gift, loan, perquisite, reward or advantage, pecuniary or otherwise, shall either directly or indirectly be given promised or offered by the contractor or any of his servants or agents through any public officer, or person in the employ of JMC / Govt. in any way relating to his office or employment, or if any such officer or person shall become in any way directly or indirectly interested in the contract the Engineer In-charge may thereupon, by notice in writing rescind the contract and the Performance Bank Guarantee (PBG) of the Contractor shall thereupon stand forfeited and be absolutely

at the disposal of JMC and the same consequences shall ensure as if the contract had been rescinded under above clause hereof; and in addition the contractor shall not be entitled to recover or be paid for any work therefore actually performed under the contract.

8.25 Payment Terms

The SI is free to modify the timelines of different sub-modules during the development & implementation period, however all modules should be covered as mentioned in the scope of work. The payment terms are to be read as percentage payment of the commercial bid section nos. (item nos.)

Category of Work	Activity/Deliverables	Timelines in Months	Percentage of Payment	Commercial Bid
Requirement Gathering	Project Kick-off meeting & presentation to departments	M1	0%	C(8,9&10)
	Inception report	M1	30%	C(8,9&10)
	FRS, SRS, Project plan	M1	0%	C(8,9&10)
Delivery & installation of Workstations and COTS GIS software	Delivery of workstations& COTS GIS Software (both desktop and Enterprise)	M2	80%	C3 (a&b)
	Installation of workstations& COTS GIS Software (both desktop and Enterprise)	M2	20%	C3 (a&b)
Satellite Imagery Procurement	Supply of Satellite Imagery	M2	100%	C1
Geo-referencing of Satellite Imagery	DGPS Survey	M2	80%	C2
	Geo-referencing of Satellite Imagery	M2	80%	C4
	Final Approval	M2	20%	C(2 & 4)
2D Base Map & Change Detection reports	2D Basemap & LULC Map	M4	80%	C5
	Change Detection reports	M4		C6
	Final Approval	M4	15%	C(5 & 6)
Underground Utility	Data Submission and	M5	100%	C7

Survey	Approval			
Development of Web Portal & Mobile Application	Development & implementation of Web GIS application for 2D& 3D City model & change detection mentioned in the scope of work	M3	50%	C8
	Development & implementation of Mobile application for change detection	M3	50%	C9
	Integration of Enterprise GIS application with existing IT & GIS application	M4	50%	C10
	Comprehensive audit including security Certification, from CERT-IN empaneled agency for all applications developed as per the scope of work	M5	10%	C(8,9&10)
	Testing of GIS application & bug fixing	M5	5%	C(8,9&10)
	UAT, Training & Go-Live	M6	5%	C(5,6,8,9 & 10)
Annual Maintenance Contract of all hardware, data, applications & services provided under the scope of work for 3 years after Go-Live	Onsite Support GIS Cell Resources (5 No's)	M7-M42	Same as Capex model as per price bid	O1
	Procurement of Fresh satellite imagery(Stereo Pair) on Yearly Interval from 2026 to 2028	M7-M42	Same as Capex model as per price bid	O2
	Change Detection reports	M7-M42		O3
	AMC of Hardware	M7-M42	Quarterly	O4
	AMC of Bug fixing, issue resolution &enhancements of the	M7-M42	Quarterly	O5

	Enterprise Geospatial applications			

8.26 Service Level Agreement

The purpose of this Service Level Requirements/Agreement (hereinafter referred to as SLR/SLA) is to clearly define the levels of service which shall be provided by the System Integrator to the Corporation for the duration of this contract period of the Project.

- Timelines specified in the above section (**Work Completion Timelines and Payment Terms**) shall form the Service Levels for delivery of Services specified there-in.
- All the payments to the System Integrator are linked to the compliance with the SLA metrics specified in this document.
- The project Service Level Agreement are proposed to be performance based. For purposes of Service Level Agreement, the definitions and terms as specified along with the following terms shall have the meanings set forth below:
 - a. “Uptime” shall mean the time period for which the specified services / components with specified technical and service standards are available for the application. Uptime, in percentage, of any component (Non-IT and IT) can be calculated as:
$$\text{Uptime} = \{1 - [(\text{System Downtime}) / (\text{Total Time} - \text{Planned Maintenance Time})]\} * 100$$
 - b. “Downtime” shall mean the time period for which the specified services / components with specified technical and service standards are not available for the Users, the scheduled outages / Planned Maintenance time planned in advance for application and related infrastructure. The planned maintenance time / scheduled downtime will include activities like software upgrades, patch management, security software installations.
 - c. The selected System Integrator will be required to schedule ‘planned maintenance time’ with prior approval of JMC. This will be planned outside working time.

Inexceptional circumstances, JMC may allow the selected System Integrator to plan scheduled downtime in the working hours.

- d. “Incident” refers to any event / abnormalities in the functioning of the application, Infrastructure and services that may lead to disruption in normal operations.
 - e. O&M Support shall mean the support for minimum 8 hrs. per day as per JMC’s working days, availability over telephone / mobile for 24 x 7 x 365 days and anytime support during exigencies and shall handle Fault reporting, Trouble Ticketing and related enquiries during this contract.
 - f. “Response Time” shall mean the time incident is reported to the “O&M Support team” and concerned expert is assigned for the issue / call.
 - g. “Resolution Time” shall mean the time taken (after the incident has been reported at the O&M Support team), in resolving (troubleshooting and fixing) or escalating (to the second level) getting the confirmatory details about the same from the selected System Integrator and conveying the same to the end user), the services related troubles during the first level escalation.
- The resolution time shall vary based on the severity of the incident reported at the O&M Support. The severity would be as follows:
 - a) **Critical / High:** The application is down impacting critical business functions or multiple modules/functions down impacting users on daily operations or any module /functionality deemed as highly critical by JMC.
 - b) **Medium:** One module/functionality down impacting critical business functions having major impact on daily operations.
 - c) **Low:** Loss of business functionality for less than 10 users impacting day to day operations or minor functionality down impacting less than 10 users.

The SLAs have been logically segregated in the following categories:

1. Pre-implementation phase & Manpower availability
2. Data & Map generation
3. Development & Implementation of Enterprise GIS application
4. Issues Resolution

5. Other Factors

6. Application Response Time Performance Criteria

Commencement of SLA: The SLA shall commence from implementation period itself for adherence to the implementation plan for new requirements. The penalty will be deducted from the next payment milestone during the implementation period.

S.No	Parameter	Metric	Base	Penalty
1	Adherence to planned implementation schedule. • Award of Contract & team mobilization • Order requiredSatellite Images • Software Requirement Specification (SRS) • Delivery of Satelliteimages	The delay for each milestone as per the planned schedule should not exceed more than a week without a justified reason agreed and approved by JMC. Thereafter penalty will be levied on daily base.	Per Day	□ 1,000 per day up to maximum 10% of contract value post which JMC may invoke annulment of the contract.
2	Mobilization of the team for commencement of work	As per the defined timeline mentioned	Per Occurrence	In case the proposed resource is not available for continuous 7 working days, the penalty of □. 10,000 per week per resource will be imposed during and after 7 days.

3	Manpower Availability & Readiness during Implementation	100% as per requirement timelines	Daily	In case the proposed resource is not available for continuous 7 working 105 days, the penalty of ₹ 10,000 per week will be imposed during and after 7 days
4	Replacement of resources during project execution	Within 15 days of release of existing resource initiated either by JMC. The resource should be of equivalent or higher qualification and experience	Per occurrence	₹ 5,000 per week of delay for deployment of equivalent or higher qualification and experienced alternate resource. In the event if selected fails to replace the resource or as per the directions given by JMC. From time to time, JMC. Reserves the right to procure similar services from market as alternate sources at the risk, cost and responsibility of. Before taking such a decision,

				JMCshall serve a notice period of 1 month
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8.27 Application Response Time Performance Criteria

Response time: Response time is the time interval between the instant at which the LAN user / Internet user at a terminal enters a request for a response from the Database server and the instant at which the response is received completely at the terminal

User Activity	LAN Users/Internet User(App Server to DB)	
	90 %of cases	All
Menu Navigation – To display the menu as per the defined user role and profile	< 4 sec	< 6 sec
Screen Opening – To display the selected data entry screen from the menu chosen	< 4 sec	< 6 sec
Field Navigation – To navigate between the data entry fields in the Screen	< 7 sec	< 10 sec
Look up response time – To display items from List of Values	< 10 sec	< 12 sec
Look up response time – To display the required GIS & map data	< 10 sec	< 15 sec
Screen Navigation – Time taken to navigate from one screen (tab page) to another which does not involve processing in earlier screen	< 5 sec	< 7 sec
Map Navigation- Time taken in map rendering with all visible layers including	< 6 sec	< 10 sec

2D/3D (with scale dependency) on using navigation tools.		
Transaction Commit – Response time taken to commit a simple transaction	< 10 sec	< 15 sec
Query Retrieval Response Time		
simple query	< 8 sec	<15 sec
Medium complexity query	< 15 sec	<20 sec
High Complexity query	<20 sec	<30 sec
Reports Generation Response Time		
simple Report	<10 sec	<20 sec
Medium Complexity report	<45 sec	<90 sec
High Complexity report	<90 sec	<3 sec

- 1) Non-adherence to the above-mentioned response time would be considered as a bug and the penalty would be levied as per the table for the Issue Resolution. Support service shall be available 24 * 7.
- 2) Bidder shall provide software to measure the performance for the above-mentioned SLAs.
- 3) The successful bidder will have to maintain strict Privacy and confidentiality of all the data it/its staff gets access to. Adequate provisions to be made to not to allow unrestricted access to the data to people in the organization who have not got the necessary permissions. Successful Bidder cannot sell or part with any data in any form. Penalty of Rs. 1,000/- to Rs. 100,000/- would be charged on default of this service condition. The exact amount of the penalty will be decided by JMC based upon severity of the default.

Applicable SLAs and Penalties for issues resolution:

The down time will be calculated on monthly bas. Non-adherence to any of the services as mentioned below will lead to penalty as per the SLA clause and will be used to calculate downtime. The downtime calculated shall not include the following:

- a) Down time due to hardware / software and application which is owned by JMC at their premises
- b) Negligence or other conduct of JMC or its agents, including a failure or malfunction resulting from applications or services provided by JMC or its vendors.
- c) Failure or malfunction of any equipment or services not provided by the Bidder. However, it is the responsibility / onus of the selected Bidder to prove that the outage is attributable to JMC. The selected Bidder shall obtain the proof authenticated by the JMC's official that the outage is attributable to the JMC.
- d) The Agency shall deploy enough manpower suitably qualified and experienced in shifts to meet the SLA. Agency shall appoint as many team members as deemed fit by them, to meet the time Schedule and SLA requirements.

Notes:

- a. The maximum penalty will be up to 10% of total project cost as per the Commercial Bid submitted by the System Integrator.
- b. The System Integrator shall deploy sufficient manpower suitably qualified and experienced to meet the SLA. The System Integrator shall appoint as many team members as deemed fit by them, to meet the time Schedule and SLA requirements.
- c. Once a maximum penalty of 10% of the contract value is reached, JMC has the right to call for the annulment of contract.

8.28 Professional Project Management

1. The selected System Integrator shall deploy resources / execute the project with complete professionalism and full commitment to the scope of work and the prescribed service levels.

System Integrator shall attend regular Project Review Meetings scheduled by JMC and shall adhere to the directions given during the meeting. Following

Responsibilities are to be executed by the System Integrator in regular manner to ensure the proper management of the project:

- Finalization of the Project plan in consultation with JMC and its consultant. Project Plan should cost of Work plan, Communication matrix, Timelines, Quality Plan, Configuration Management Plan.
- Plan and deploy the resources in conjunction with the Project Plan and to execute roles and Responsibilities against each activity of the project plan
- Preparation and regular update of the Risk Register and the Mitigation Plan. Timely communication of the same to all the identified project stakeholders
- Submission of Weekly Project Progress Reports
- Monthly Compliance report, which will cover compliances to Project Timelines, Project Team, Hardware (if applicable) and Software delivered, SLAs.

8.29 Use and Acquisition of Assets during the term

The selected System Integrator shall:

- A. Take all reasonable and proper care of the entire hardware and software, network or any other information technology infrastructure components used during the project and other facilities leased/owned by the System Integrator exclusively in terms of the delivery of the services as per this CA (hereinafter the “Assets”) in proportion to their use and control of such Assets which will include all upgrades/enhancements and improvements to meet the needs of the project arising from time to time
- B. Term “Assets” also refers to all the hardware / Software / furniture / data / documentations / manuals / catalogues / brochures / or any other material procured, created or utilized by the System Integrator.
- C. Keep all the tangible Assets in good and serviceable condition (reasonable wear and tear excepted) suitably upgraded subject to the relevant standards as stated in the bid to meet the SLAs mentioned in the contract and during the entire term of the Agreement
- D. Ensure that any instructions or manuals supplied by the manufacturer of the Assets for use of Assets and which are provided to the System Integrator will be

followed by the System Integrator and any person who will be responsible for the use of the Asset

- E. Take such steps as may be recommended by the manufacturer of the Assets and notified to the System Integrator or as may be necessary to use the Assets in a safe manner
- F. To the extent that the Assets are under the control of the System Integrator, keep the Assets suitably housed and in conformity with any statutory requirements from time to time applicable to them
- G. Not, knowingly, or negligently use or permit any of the Assets to be used in contravention of any statutory provisions or regulation or in any way contrary to law
- H. Use the Assets exclusively for the purpose of providing the Services as defined in the contract
- I. Ensure the integration of the software with hardware to be setup and the current Assets in order to ensure the smooth operations of the entire solution architecture to provide efficient services to JMC of this Project in an efficient and speedy manner
- J. System Integrator shall not use JMC data to provide services for the benefit of any third party, as a service bureau or in any other manner

8.30 Security and safety

- A. The System Integrator will comply with the directions issued from time to time by JMC and the standards related to the security and safety as far as it applies to the provision of the Services
- B. System Integrator shall also comply with JMC / Government of Gujarat's / Government of India's information technology security and standard policies in force from time to time as applicable. JMC shall share the relevant guidelines and standards to the System Integrator upon signing of the CA.
- C. System Integrator shall use reasonable endeavors to report forthwith in writing to all the partners / contractors about the civil and criminal liabilities accruing due to

any unauthorized access (including unauthorized persons who are employees of any Party) or interference with JMC's data, facilities, or Confidential Information.

- D. The System Integrator shall upon reasonable request by JMC or his/her nominee(s) participate in regular meetings when safety and information technology security matters are reviewed.
- E. System Integrator shall promptly report in writing to JMC any act or omission which they are aware that could have an adverse effect on the proper conduct of safety and information technology security at JMC.

8.31 Indemnity

- A. The System Integrator agrees to indemnify and hold harmless JMC, its officers, employees and agents (each an "Indemnified Party") promptly upon demand at any time and from time to time, from and against any and all losses, claims, damages, liabilities, costs (including reasonable attorney's fees and disbursements) and expenses (collectively, "Losses") to which the Indemnified Party may become subject, in so far as such losses directly arise out of, in any way relate to, or result from:
- B. Any miss-statement or any breach of any representation or warranty made by the System Integrator or
- C. The failure by the System Integrator to fulfill any covenant or condition contained in this Agreement, including without limitation the breach of any terms and conditions of this Agreement by any employee or agent of the System Integrator. Against all losses or damages arising from claims by third Parties that any Deliverable (or the access, use or other rights thereto), created by System Integrator pursuant to this Agreement, or any equipment, software, information, methods of operation or other intellectual property created by System Integrator or sub-contractors pursuant to this Agreement, or the SLAs:
 - i) infringes a copyright, trade mark, trade Design enforceable in India,
 - ii) infringes a patent issued in India, or
 - iii) constitutes misappropriation or unlawful disclosure or use of another Party's trade secrets under the laws of India (collectively, "Infringement Claims");

provided, however, that this will not apply to any Deliverable (or the access, use or other rights thereto) created by: (A) to Offer GIS based Services in JMC by itself or through other persons other than System Integrator or its sub-contractors; (B) Third Parties (i.e., other than System Integrator or sub-contractors) at the direction of JMC, or

- D. Any compensation / claim or proceeding by any third party against JMC arising out of any act, deed or omission by the System Integrator or
- E. Claim filed by a workman or employee engaged by the System Integrator for carrying out work related to this Agreement. For the avoidance of doubt, indemnification of Losses pursuant to this section shall be made in an amount or amounts enough to restore each of the Indemnified Party to the financial position it would have been in had the losses not occurred.
- F. Any payment made under this Agreement to an indemnity or claim for breach of any provision of this Agreement shall include applicable taxes

8.32 Third Party Claims

- A. Subject to Sub-clause (2.) below, the System Integrator (the “Indemnified Party”) from and against all losses, claims litigation and damages on account of bodily injury, death or damage to tangible personal property arising in favor or any person, corporation or other entity (including the Indemnified Party) attributable to the Indemnifying Party’s performance or non-performance under this Agreement or the SLAs.
- B. The indemnities set out in Sub-clause (1.) above shall be subject to the following conditions:
 - a) The Indemnified Party, as promptly as practicable, informs the Indemnifying Party in writing of the claim or proceedings and provides all relevant evidence, documentary or otherwise;
 - b) The Indemnified Party shall, at the cost and expenses of the Indemnifying Party, give the Indemnifying Party all reasonable assistance in the defense of such claim including reasonable access to all relevant information, documentation, and personnel. The indemnifying party shall bear cost and

expenses and fees of the Attorney on behalf of the Indemnified Party in the litigation, claim.

- c) If the Indemnifying Party does not assume full control over the defense of a claim as provided in this Article, the Indemnifying Party may participate in such defense at its sole cost and expense, and the Indemnified Party will have the right to defend the claim in such manner as it may deem appropriate, and the cost and expense of the Indemnified Party will be borne and paid by the Indemnifying Party.
- d) The Indemnified Party shall not prejudice, pay or accept any proceedings or claim, or compromise any proceedings or claim, without the written consent of the Indemnifying Party.
- e) System Integrator hereby indemnify and hold indemnified JMC harmless from and against any and all damages, losses, liabilities, expenses including legal fees and cost of litigation in connection with any action, claim, suit, proceedings as if result of claim made by the third party directly or indirectly arising out of or in connection with this agreement.
- f) All settlements of claims subject to indemnification under this Article will: (a) be entered into only with the consent of the Indemnified Party, which consent will not be unreasonably withheld and include an unconditional release to the Indemnified Party from the claimant for all liability in respect of such claim; and (b) include any appropriate confidentiality agreement prohibiting disclosure of the terms of such settlement.
- g) The Indemnified Party shall take steps that the Indemnifying Party may reasonably require to mitigate or reduce its loss as a result of such a claim or proceedings and
- h) In the event that the Indemnifying Party is obligated to indemnify an Indemnified Party pursuant to this Article, the Indemnifying Party will, upon payment of such indemnity in full, be subrogated to all rights and defenses of the Indemnified Party with respect to the claims to which such indemnification relates.

- i) In the event that the Indemnifying Party is obligated to indemnify the Indemnified Party pursuant to this Article, the Indemnified Party will be entitled to invoke the Performance Bank Guarantee, if such indemnity is not paid, either in full or in part, and on the invocation of the Performance Bank Guarantee, the Indemnifying Party shall be subrogated to all rights and defenses of the Indemnified Party with respect to the claims to which such indemnification relates.

System Integrator will defend or settle third party claims against JMC solely attributable to the System Integrator's infringement of any copyrights, trademarks or industrial Design rights alleged to have occurred in respect of System Integrator branded hardware/software/deliverables etc. (together "deliverables") supplied by the System Integrator.

The System Integrator shall pay all costs, damages and attorney's fees that a court finally awards.

JMC shall provide the System Integrator with prompt notice of such claim and extend full cooperation and assistance, information and authority reasonably necessary to defend or settle such claim. The System Integrator will have adequate opportunity to control the response thereto and the defense thereof.

Further as an exclusion, the System Integrator shall have no obligation for any claim of infringement to the extent arising from use of the deliverables in a way not indicated in the statement of work or in any specifications or documentation provided with such deliverable.

8.33 Publicity

Any publicity by the System Integrator in which the name of JMC is to be used should be done with the explicit written permission of The Municipal Commissioner, Assistant Commissioner, Jamnagar Municipal Corporation.

8.34 Warranties

- 1) The System Integrator warrants and represents to JMC that:
 - i) It has full capacity and authority and all necessary approvals to enter into and to perform its obligations under this Agreement.

- ii) This Agreement is executed by a duly authorized representative of the System Integrator.
 - iii) It shall discharge its obligations under this Agreement with due skill, care and diligence so as to comply with the service level agreement.
- 2) In the case of the SLAs, the System Integrator warrants and represents to JMC, that:
- i) The System Integrator has full capacity and authority and all necessary approvals to enter into and perform its obligations under the SLAs and to provide the Services
 - ii) The SLAs shall be executed by a duly authorized representative of the System Integrator
 - iii) The Services will be provided and rendered by appropriately qualified, trained and experienced personnel as mentioned in the bid
 - iv) System Integrator has and will have all necessary licenses, approvals, consents of third Parties free from any encumbrances and all necessary technology, hardware and software to enable it to provide the Services
 - v) The Services will be supplied in conformance with all laws, enactments, orders and regulations applicable from time to time
 - vi) System Integrator will warrant that the solution provided under the contract is new, of the most recent higher Version /models and incorporate all recent improvements in Design and materials unless provided otherwise in the contract
 - vii) The System Integrator shall ensure defect free operation of the entire solution and shall replace any such components, equipment's, software and hardware (if applicable) which are found defective and during the entire contract period the System Integrator shall apply all the latest upgrades / patches / releases for the software after appropriate testing. No additional costs shall be paid separately for the warranty other than what are the costs quoted by the System Integrator and as specified in the contract.

viii) If the System Integrator uses in the course of the provision of the Services, components, equipment's, software and hardware manufactured by any third party and which are embedded in the Deliverables or are essential for the successful use of the Deliverables, it will pass-through third-party manufacturer's Warranties relating to those components, equipment, software and hardware to the extent possible.

The System Integrator will repair/correct the warranty defect in the System Integrator branded hardware/software, or correct a material non-conformance to specifications in accordance with the terms and conditions mutually agreed between the parties during the agreed warranty period and in accordance with the terms of the respective OEMs/Software vendors. Further, System Integrator will not be responsible for any breach of warranty or support resulting from unauthorized changes made to the software/hardware supplied by it.

Notwithstanding what has been stated elsewhere in this Agreement and the Schedules attached herein, in the event the System Integrator is unable to meet the obligations pursuant to the Implementation of the project, Operations and Maintenance Services and any related scope of work as stated in this Agreement and the Schedules attached herein, JMC will have the option to invoke the Performance Guarantee after serving a written notice of thirty (30) days to the System Integrator.

8.35 Force Majeure

The System Integrator shall not be liable for forfeiture of its Performance Guarantee, imputation of liquidated damages or termination for default, if and to the extent that it's delay in performance or other failure to perform its obligations under the contract is the result of an event of Force Majeure. For purposes of this Clause, "Force Majeure" means an event beyond the "reasonable" control of the System Integrator, not involving the System Integrator's fault or negligence and not foreseeable. Unforeseen circumstances or causes beyond the control of the System Integrator include but are not limited to acts of God, war, riot, acts of civil or military authorities, fire, floods, accidents, terrorist activity, Epidemic, Pandemic, strikes or shortages of transportation facilities, fuel, energy, labor or material.

For the System Integrator to take benefit of this clause it is a condition precedent that the System Integrator must promptly notify JMC, in writing of such conditions and

the cause thereof within five calendar days of the arising of the Force Majeure event. JMC, or the consultant / committee appointed by JMC shall study the submission of the System Integrator and inform whether the situation can be qualified one of Force Majeure. Unless otherwise directed by JMC in writing, the System Integrator shall continue to perform its obligations under the resultant Agreement as far as it is reasonably practical, and shall seek all reasonable alternative means for performance of services not prevented by the existence of a Force Majeure event.

In the event of delay in performance attributable to the presence of a force majeure event, the time for performance shall be extended by a period(s) equivalent to the duration of such delay.

If the duration of delay continues beyond a period of 30 days, JMC and the System Integrator shall hold consultations with each other in an endeavor to find a solution to the problem.

Notwithstanding anything to the contrary mentioned above, the decision of JMC shall be final and binding on the System Integrator.

8.36 Resolution of Disputes

JMC and the System Integrator shall make every attempt to resolve dispute amicably, by direct information, negotiations of any disagreement or dispute arising between them under or in connection with this Agreement. All differences disputes arising under and out of these present, or in connection with this Agreement shall be referred to the Municipal Commissioner of JMC and the decision of the Municipal Commissioner shall be final and binding on both the Parties.

8.37 Risk Purchase Clause

In the event, selected System Integrator fails to execute the project as stipulated in the CA, or as per the directions given by JMC from time to time, JMC reserves the right to procure similar services from the next selected / eligible Bidder or from alternate sources at the cost of the System Integrator. Before taking such a decision, JMC shall serve a notice period of one month to the System Integrator.

The 30 days' notice period shall be considered as the 'Cure Period' to facilitate the System Integrator to cure the breach. The provision for Risk Purchase shall be evoked in the event the System Integrator fails to correct the breach within the 'Cure Period'.

Further, the System Integrator's liability to pay shall be set as 25% of the value of the undelivered services.

8.38 Limitation of Liability towards JMC

The System Integrator's liability under the resultant Agreement shall be determined as per the Law in force for the time being. The System Integrator shall be liable to JMC for loss or damage occurred or caused or likely to occur on account of any act of omission on the part of the System Integrator and its employees, including loss caused to JMC on account of defect in goods or deficiency in services on the part of System Integrator or his agents or any person / persons claiming through or under said System Integrator.

selected System Integrator's aggregate liability in connection with obligations undertaken as a part of this contract regardless of the form or nature of the action giving rise to such liability, shall be at actual and limited to the amount paid by JMC for:

i) services provided during the twelve (12) months immediately preceding the date of the claim that in each case is the subject of the claim.

This limit shall not apply to damages for bodily injury (including death) and damage to real property and tangible personal property for which the System Integrator is legally liable.

8.39 Conflict of Interest

The System Integrator shall disclose to JMC in writing, all actual and potential conflicts of interest that exist, arise or may arise (either for the System Integrator or its team) in the course of performing the Services as soon as it becomes aware of such a conflict. System Integrator shall hold JMC's interest paramount, without any consideration for future work, and strictly avoid conflict of interest with other assignments.

8.40 Data Ownership

All the data created within the scope of this contract shall be owned by JMC. The System Integrator shall take utmost care in maintaining security, confidentiality and backup of this data. Access to the data / systems shall be given by the System Integrator only to the personnel working on the projects and their names and contact

details shall be shared with JMC in advance. JMC / its authorized representative(s) shall conduct periodic / surprise security reviews and audits, to ensure the compliance by the System Integrator to data / system security.

The ownership of the application and the data shall rest with JMC.

8.41 Fraud and Corruption

JMC requires that System Integrator must observe the highest standards of ethics during the execution of the contract. In pursuance of this policy, JMC defines, for the purpose of this provision, the terms set forth as follows:

i) “Corrupt practice” means the offering, giving, receiving or soliciting of anything of value to influence the action of JMC in contract executions.

ii) “Fraudulent practice” means a misrepresentation of facts, in order to influence a procurement process or the execution of a contract, to JMC, and includes collusive practice among bidders (prior to or after bid submission) designed to establish bid prices at artificially high or non-competitive levels and to deprive JMC of the benefits of free and open competition.

iii) “Undesirable practice” means:

(a) establishing contact with any person connected with or employed or engaged by JMC with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Selection Process; or

(b) Having a Conflict of Interest; and

iv) “Restrictive practice” means forming a cartel or arriving at any understanding or arrangement among Bidders with the objective of restricting or manipulating a full and fair competition in the Selection Process.

v) “Coercive Practices” means harming or threatening to harm, directly or indirectly, persons or their property to influence their participation in the execution of contract.

If it is noticed that the System Integrator has indulged into the Corrupt / Fraudulent / Undesirable / Coercive practices, it will be an enough ground for JMC for termination of the contract and initiate black-listing of the vendor.

8.42 Exit Management

A. Exit Management Purpose

This clause sets out the provision, which will apply during Exit Management period. The Parties shall ensure that their respective associated entities carry out their respective obligations set out in this Exit Management Clause.

The exit management period starts, in case of expiry of contract, at least 6 months prior to the date when the contract comes to an end or in case of termination of contract, on the date when the notice of termination is sent to the System Integrator. The exit management period ends on the date agreed upon by JMC or three months after the beginning of the exit management period, whichever is earlier.

B. Confidential Information, Security and Data System Integrator will promptly, on the commencement of the exit management period, supply to JMC or its nominated agency the following:

- a. Information relating to the current services rendered and performance data relating to the performance of the services; documentation relating to the project, project's customized source code; any other data and confidential information created as part of or is related to this project;
- b. Project data as is reasonably required for purposes of the project or for transitioning of the services to its replacing successful Bidder in a readily available format.
- c. All other information (including but not limited to documents, records and agreements) relating to the services reasonably necessary to enable JMC and its nominated agency, or its replacing vendor to carry out due diligence in order to transition the provision of the Services to JMC or its nominated agency, or its replacing vendor (as the case maybe).
- d. The System Integrator shall retain all of the above information with them for 30 days after the termination of the contract, post which the provider has to wipe/purge/delete all information created or retained as part of this project.
- e. System Integrator will sign a Non-Disclosure Agreement with JMC RE Department. The format for the same has been included in Annexure VII.

C. Rights of Access to Information

At any time during the exit management period, the System Integrator will be obliged to provide an access of information to JMC and / or any Replacing Vendor in order to make an inventory of the Assets (including Hardware / Software / Active / passive), documentations, manuals, catalogs, archive data, Live data, policy documents or any other material related to implementation of the application for JMC.

D. Exit Management Plan

Successful selected agency shall provide JMC with a recommended “Exit Management Plan” within 90 days of signing of the contract, which shall deal with at least the following aspects of exit management in relation to the SLA as a whole and in relation to the Project Implementation, the Operation and Management SLA and Scope of work definition.

- a. A detailed program of the transfer process that could be used in conjunction with a Replacement Vendor including details of the means to be used to ensure continuing provision of the services throughout the transfer process or until the cessation of the services and of the management structure to be used during the transfer
- b. Plans for the communication with such of the successful bidder, staff, suppliers, customers and any related third party as are necessary to avoid any material detrimental impact on Project’s operations as a result of undertaking the transfer
- c. Plans for provision of contingent support to the implementation of hosted Project and Replacement Vendor for a reasonable period (minimum one month) after transfer
- d. Exit Management Plan shall be presented by the System Integrator to and approved by JMC or its nominated agency
- e. The terms of payment as stated in the Terms of Payment Schedule include the costs of the System Integrator complying with its obligations under this Schedule
- f. During the exit management period, the System Integrator shall use its best efforts to deliver the services

g. Payments during the Exit Management period shall be made in accordance with the Terms of Payment Schedule

8.43 Termination of contract

JMC may, without prejudice to any other remedy under this Contract and applicable law, reserves the right to terminate for breach of contract by providing a written notice of 30 days stating the reason for default to the System Integrator and terminate the contract either in whole or in part:

- If the System Integrator fails to deliver any or all of the project requirements / operationalization / Operational Acceptance of project within the time frame specified in the contract, or
- If the System Integrator fails to perform any other obligation(s) under the contract.

Prior to providing a notice of termination to the System Integrator, JMC shall provide the System Integrator with a written notice of 30 days instructing the System Integrator to cure any breach/ default of the Contract, if JMC is of the view that the breach may be rectified.

On failure of the System Integrator to rectify such breach within 30 days, JMC may terminate the contract by providing a written notice of 30 days to the System Integrator, provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to JMC. In such an event the System Integrator shall be liable for penalty imposed by JMC.

In the event of termination of this contract for any reason whatsoever, JMC is entitled to impose any such obligations and conditions and issue any clarifications as may be necessary to ensure an efficient transition and effective continuity of the services which the System Integrator shall be obliged to comply with and take all available steps to minimize the loss resulting from that termination/ breach, and further allow and provide all such assistance to JMC and/ or succeeding vendor, as may be required, to take over the obligations of the System Integrator in relation to the execution /continued execution of the requirements of this contract.

8.44 Miscellaneous

a. Confidentiality

“Confidential Information” means all information including Project Data (whether in written, oral, electronic or other format) which relates to the technical, financial and operational affairs, business rules, citizen information, Design rights, know-how and personnel of each Party and its affiliates which is disclosed to or otherwise learned by the other Party or its subcontractors (whether a Party to the contract or to the SLA) in the course of or in connection with the contract (including without limitation such information received during negotiations, location visits and meetings in connection with the contract or to the SLA) or pursuant to the contract to be signed subsequently.

Except with the prior written permission of JMC, the System Integrator (including all partners) and its Personnel shall not disclose such confidential information to any person or entity not expected to know such information by default of being associated with the project, nor shall the System Integrator and its Personnel make public the recommendations formulated in the course of, or as a result of the project. In matters pertaining to privacy of data, the System Integrator (including all partners) shall not use any data for analytical/commercial reasons whatsoever.

The System Integrator recognizes that during the term of this Agreement, sensitive data will be procured and made available to it, its Sub contractors and agents and others working for or under the System Integrator. Disclosure or usage of the data by any such recipient may constitute a breach of law applicable causing harm not only to the Department whose data is used but also to its stakeholders. The function of JMC requires the System Integrator, its Subcontractors, and agents to demonstrate utmost care, sensitivity, and strict confidentiality. Any breach of this Article will result in JMC and its nominees receiving a right to seek injunctive relief and damages, from the System Integrator.

The restrictions of this Article shall not apply to confidential information that:

- i. Is or becomes generally available to the public through no breach of this Article by the Recipient; and
- ii. Was in the recipient’s possession free of any obligation of confidence prior to the time of receipt of it by the Recipient hereunder; and
- iii. Is developed by the Recipient independently of any of discloser’s Confidential Information; and

- iv. Is rightfully obtained by the Recipient from third Parties authorized at that time to make such disclosure without restriction; and
- v. Is identified in writing by the Discloser as no longer proprietary or confidential; or vi. Is required to be disclosed by law, regulation or Court Order, provided that the recipient gives prompt written notice to the Discloser of such legal and regulatory requirement to disclose so as to allow the Discloser reasonable opportunity to consists such disclosure.

To the extent that such disclosure is required for the purposes of this Agreement, either Party may disclose Confidential Information to:

- i. Its employees, agents, and independent contractors and to any of its affiliates and their respective independent contractors or employees; and
- ii. Its professional advisors and auditors, who require access for the purposes of this Agreement, whom the relevant Party has informed of its obligations under this Article and in respect of whom the relevant Party has informed of its obligations under this Article has used commercially reasonable efforts to ensure that they are contractually obliged to keep such Confidential Information confidential on terms substantially the same as set forth in this Article. Either Party may also disclose confidential Information or any entity with the other Party's prior written consent.

The provisions of this Article shall survive the expiration or any earlier termination of this Agreement.

b. Standards of Performance

The System Integrator shall provide the services and carry out their obligations under the Contract with due diligence, efficiency and professionalism/ethics in accordance with generally accepted professional standards and practices. The System Integrator shall always act in respect of any matter relating to this contract. The System Integrator shall abide by all the applicable Provisions / Acts / Rules / Regulations, Standing orders, etc. of Information Technology as prevalent in the country. The

System Integrator shall also conform to the standards laid down by Government of Maharashtra or Government of India from time to time. Such standards and guidelines shall be shared with the System Integrator by JMC up on signing of the Contract.

c. Sub Contracts

Sub-contracting is not permitted within the scope of this tender.

d. Care to be taken while working at JMC Office

System Integrator should follow instructions issued by concerned Competent Authority from time to time for carrying out work at Designated places. System Integrator should ensure that there is no damage caused to any private or public property. In case such damage is caused, System Integrator shall immediately bring it to the notice of concerned organization and JMC in writing and pay necessary charges towards fixing of the damage. System Integrator shall ensure that its

employees/representatives don't breach privacy of any citizen or establishment during the course of execution or maintenance of the project.

e. Compliance with Labour regulations

The System Integrator shall pay fair and reasonable wages to the workmen employed, for the contract undertaken and comply with the provisions set forth under the Minimum wages Act and the Contract Labour Act 1970. The salary of the manpower working on JMC project should be paid using ECS / NEFT / RTGS. A record of the payments made in this regard should be maintained by the System Integrator. Upon request, this record shall be produced to the appropriate authority in JMC and/or Judicial Body. If complaints are received by JMC (or any appropriate authority) appropriate action (Liquidation of Security Deposit, Blacklisting, etc.) may be initiated as deemed necessary against the System Integrator. The employees of the System Integrator in no case shall be treated as

the employees of the JMC at any point of time. Since the employee working on project are employee s of the System Integrator, the payment of salary/ compensation if any shall be entire responsibility of. Under no circumstances the JMC shall be liable to pay the same.

f. Independent Contractor

Nothing in this Agreement shall be construed as establishing or implying any partnership or joint venture or employment relationship between the Parties to this Agreement. Except as expressly stated in this Agreement nothing in this Agreement shall be deemed to constitute any Party as the agent of any other Party or authorizes either Party:

- (i) to incur any expenses on behalf of the other Party,
- (ii) to enter into any engagement or make any representation or warranty on behalf of the other Party,
- (iii) to pledge the credit of or otherwise bind or oblige the other Party, or
- (iv) to commit the other Party in any manner whatsoever in each case without obtaining the other Party's prior written consent.

g. Waiver

A waiver of any provision or breach of this Agreement must be in writing and signed by an authorized official of the Party executing the same. No such waiver shall be construed to affect or imply a subsequent waiver of the same provision or subsequent breach of this Agreement.

h. Notices

Any notice or other document, which may be given by either Party under this Agreement, shall be given in writing in person or by pre-paid recorded delivery post. In relation to a notice given under this Agreement, any such notice or other document shall be addressed to the other Party's principal or registered office address as set out below:

JMC:

Municipal Commissioner,

Jamnagar Municipal Corporation, Municipal Head Office,

Jamnagar

Tel:

System Integrator:

Tel:

Any notice or other document shall be deemed to have been given to the other Party when delivered (if delivered in person) between the hours of 9.30 A.M and 5.30 P.M. at the address of the other Party set forth above or on the next working day thereafter if delivered outside such hours, and seven calendar days from the date of posting (if by letter).

i. Personnel/Employees

i. Personnel/employees assigned by selected System Integrator to perform the services shall be employees of System Integrator, under no circumstances will such personnel be considered as employees of JMC. selected System Integrator shall have the sole responsibility for supervision and control of its personnel and for payment of such personnel's employee's entire compensation, including salary, legal deductions withholding of income taxes and social security taxes, worker's compensation, employee and disability benefits and the like and shall be responsible for all employer obligations under all laws as applicable from time to time. JMC shall not be responsible for the above issues concerning to personnel of System Integrator.

ii. selected System Integrator shall use its best efforts to ensure that enough System Integrator personnel are employed to perform the Services, and that, such personnel have appropriate qualifications to perform the Services. JMC or its nominated agency shall have the right to require the removal or replacement of any System Integrator personnel performing work under this Agreement. In the event that JMC requests that any System Integrator personnel be replaced, the substitution of such personnel shall be accomplished pursuant to a mutually agreed upon schedule and upon clearance of the personnel based on profile review and personal interview by JMC or its nominated agency as per defined SLAs. The System Integrator shall depute quality team for the project and as per requirements JMC shall have the right to ask System Integrator to change the team.

iii. Management (Regional Head / VP level officer) of System Integrator needs to be involved in the project monitoring and should attend the review meeting at least once in a month.

iv. The profiles of resources proposed by System Integrator in the technical bid, which are considered for Technical bid evaluation, shall be construed as 'Key Personnel' and the SystemIntegrator shall not remove such personnel without the prior written consent of JMC. For any changes to the proposed resources, System Integrator shall provide equivalent or more experienced resources in consultation with JMC.

v. Except as stated in this clause, nothing in this Agreement will limit the ability of System Integrator freely to assign or reassign its employees; provided that System Integrator shall be responsible, at its expense, for transferring all appropriate knowledge from personnel being replaced to their replacements. JMC shall have the right to review and approve System Integrator's plan for any such knowledge transfer. System Integrator shall maintain the same standards for skills and professionalism among replacement personnel as in personnel being replaced.

vi. Each Party shall be responsible for the performance of all its obligations under this Agreement and shall be liable for the acts and omissions of its employees and agents in connection therewith.

j. Variations and Further Assurance

a. No amendment, variation or other change to this Agreement or the SLAs shall be valid unless made in writing and signed by the duly authorized representatives of the Parties to this Agreement.

b. Each Party to this Agreement or the SLAs agree to enter into or execute, without limitation, whatever other agreement, document, consent and waiver and to do all other things which shall or may be reasonably required to complete and deliver the obligations set out in the Agreement or the SLAs.

k. Survivability and Waiver

I. If any provision of this Agreement or the SLAs, or any part thereof, shall be found by any court or administrative body of competent jurisdiction to be illegal, invalid or unenforceable the illegality, invalidity or unenforceability of such

provision or part provision shall not affect the other “Corrupt practice” means the offering, giving, receiving or soliciting of anything of value to influence the action of JMC in contract executions.

II. “Fraudulent practice” means a misrepresentation of facts, in order to influence a procurement process or the execution of a contract, to JMC, and includes collusive practice among bidders (prior to or after bid submission) designed to establish bid prices at artificially high or non-competitive levels and to deprive JMC of the benefits of free and open competition.

III. “Undesirable practice” means:

- a. establishing contact with any person connected with or employed or engaged by JMC with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Selection Process; or
- b. having a Conflict of Interest; and

IV. “Restrictive practice” means forming a cartel or arriving at any understanding or arrangement among Bidders with the objective of restricting or manipulating a full and fair competition in the Selection Process.

V. “Coercive Practices” means harming or threatening to harm, directly or indirectly, persons or their property to influence their participation in the execution of contract.

If it is noticed that the System Integrator has indulged into the Corrupt / Fraudulent / Undesirable / Coercive practices, it will be an enough ground for JMC for termination of the contract and initiate black-listing of the vendor.

- i. of this Agreement or the SLAs or the remainder of the provisions in question which shall remain in full force and effect. The relevant Parties shall negotiate in good faith in order to agree to substitute for any illegal, invalid or unenforceable provision a valid and enforceable provision which achieves to the greatest extent possible the economic, legal and commercial objectives of the illegal, invalid or unenforceable provision or part provision within seven working days.

- ii. No failure to exercise or enforce and no delay in exercising or enforcing on the part of either Party to this Agreement or the SLAs of any right, remedy or provision of this Agreement or the SLAs shall operate as a waiver of such right, remedy or provision in any future application nor shall any single or partial exercise or enforcement of any right, remedy or provision preclude any other or further exercise or enforcement of any other right, remedy or provision.

I. Survivability

The termination or expiry of this Agreement or the SLAs for any reason shall not affect or prejudice any terms of this Agreement, or the rights of the Parties under them which are either expressly or by implication intended to come into effect or continue in effect after such expiry or termination.

8.45 Bank Guarantee

The System Integrator shall furnish a Bank Guarantee of Rs.----- /- (Rupees only) on the execution of these presents as Security Depot for due performance of the terms and conditions of this Contract Agreement and the same shall remain with the JMC as a depot until the expiry of the term of this Contract Agreement. This Bank Guarantee shall be forfeited by the JMC, in the event of the System Integrator committing any breach in the observance and performance of these terms and conditions and this Contract Agreement which shall then stand terminated automatically. If the System Integrator shall have duly and faithfully observed the terms and conditions of this Contract Agreement and have satisfied all claims properly chargeable against them hereunder up to the date of completion of Contract Agreement, the said Bank Guarantee will be released to the System Integrator. The System Integrator shall not be entitled to any interest on the said Bank Guarantee. If the PBG is not submitted within the stipulated time a penalty of Rs 5000/- per day will be applicable to the Bidder.

8.46 Applicable Law

The contract shall be governed by the laws and procedures prescribed by the Laws prevailing and in force in India, within the framework of applicable legislation and enactment made from time to time concerning such commercial dealings/processing. All legal disputes are subject to the jurisdiction of Mumbai courts only.

8.47 Jurisdiction of Courts

In case of any claim, dispute or difference arising in respect of this contract, the cause of action thereof shall be deemed to have arisen in Jamnagar and all legal proceedings in respect of any such claim, dispute or difference shall be referred to the Competent Court in the City of Jamnagar.

9 Annexure V: List of Approved Banks

The Earnest Money Deposit (EMD) / Performance Bank Guarantee (PBG) issued by any nationalized or Scheduled Bank having branches at Jamnagar.

10 Annexure VI: Authorization letter for attending Pre-Bid meeting / bid opening

(to be provided on the letter head of SI)

No.....

Date.....

To

The.....

Jamnagar Municipal Corporation,

Jamnagar.

Sub: Bid No..... due date.....

Sir,

We here by authorize Mr. / Ms.as our authorized representative, to represent us on the following occasion: -

- Pre-bid Meeting to be held on.....at.....A.M./P.M.
- Bid Opening on..... At..... A.M. /P.M.

Kindly permit him/her to attend the same.

Yours faithfully,

Signature:

Name of signatory:

Designation:

Rubber Stamp:

11 Annexure VII: Draft Non-Disclosure Agreement

(To be submitted on a ☐ 100 Stamp Paper)

This Non-Disclosure Agreement (“Non-Disc”) is made and entered into ____ day of ____ month ____ year (effective date) by and between _____ (“Department”) and _____ (“Company”).

Whereas, Department and Company have entered into an Agreement (“Agreement”) _____ effective _____ for _____;
and

Whereas, Each party desires to disclose to the other party certain information in oral or written form which is proprietary and confidential to the disclosing party, (“CONFIDENTIAL INFORMATION”).

NOW, THEREFORE, in consideration of the foregoing and the covenants and agreements contained herein, the parties agree as follows:

1. Definitions. As used herein:

a) The term “Confidential Information” shall include, without limitation, all information and materials, furnished by either Party to the other in connection with citizen/users/persons/customers data, products and/or services, including information transmitted in writing, orally, visually, (e.g. video terminal display) or on magnetic or optical media, and including all proprietary information, customer and prospect lists, trade secrets, trade names or proposed trade names, methods and procedures of operation, commercial or marketing plans, licensed document know-how, ideas, concepts, designs, drawings, flow charts, diagrams, quality manuals, checklists, guidelines, processes, formulae, source code materials, specifications, programs, software packages, codes and other intellectual property relating to the disclosing party’s data, computer database, products and/or services. Results of any tests, sample surveys, analytics, data mining exercises or usages etc. carried out by the receiving party in connection with the Department’s information including citizen/users/persons/customers personal or sensitive personal information as defined

under any law for the time being in force shall also be considered Confidential Information.

b) The term, “Department” shall include the officers, employees, agents, consultants, contractors and representatives of Department.

c) The term, “Company” shall include the directors, officers, employees, agents, consultants, contractors and representatives of Company, including its applicable affiliates and subsidiary companies.

2. Protection of Confidential Information: With respect to any Confidential Information disclosed to it or to which it has access, Company affirms that it shall:

a) Use the Confidential Information as necessary only in connection with Project and in accordance with the terms and conditions contained herein;

b) Maintain the Confidential Information in strict confidence and take all reasonable steps to enforce the confidentiality obligations imposed hereunder, but in no event take less care with the Confidential Information than the parties take to protect the confidentiality of its own proprietary and confidential information and that of its clients;

c) Not to make or retain copy of any commercial or marketing plans, citizen/users/persons/customers database, Bids developed by or originating from Department or any of the prospective clients of Department except as necessary, under prior written intimation from Department, in connection with the Project, and ensure that any such copy is immediately returned to Department even without express demand from Department to do so;

d) Not disclose or in any way assist or permit the disclosure of any Confidential Information to any other person or entity without the express written consent of the other party; and

e) Return to the other party, or destroy, at Department’s discretion, any and all Confidential Information disclosed in a printed form or other permanent record, or in any other tangible form (including without limitation, all copies, notes, extracts, analyses, studies, summaries, records and reproductions thereof) immediately upon

the earlier to occur of (i) expiration or termination of either party's engagement in the Project, or (ii) the request of the other party therefore.

f) Not to discuss with any member of public, media, press, any or any other person about the nature of arrangement entered between Department and Company or the nature of services to be provided by the Company to the Department.

3. Onus. Company shall have the burden of proving that any disclosure or use inconsistent with the terms and conditions hereof falls within any of the foregoing exceptions.

4. Exceptions. These restrictions as enumerated in section 1 of this Agreement shall not apply to any Confidential Information:

a) Which is independently developed by Company or lawfully received from another source free of restriction and without breach of this Agreement; or

b) After it has become generally available to the public without breach of this Agreement by Company; or

c) Which at the time of disclosure to Company was known to such party free of restriction and evidenced by documentation in such party's possession; or

d) Which Department agrees in writing is free of such restrictions.

e) Which is received from a third party not subject to the obligation of confidentiality with respect to such Information;

5. Remedies. Company acknowledges that:

a) Any actual or threatened disclosure or use of the Confidential Information by Company would be a breach of this agreement and may cause immediate and irreparable harm to Department;

b) Company affirms that damages from such disclosure or use by it may be impossible to measure accurately; and

c) Injury sustained by Department may be impossible to calculate and remedy fully. Therefore, Company acknowledges that in the event of such a breach, Department shall be entitled to specific performance by Company of Company's obligations contained in this Agreement. In addition Company shall indemnify Department of the

actual and liquidated damages which may be demanded by Department. Moreover, Department shall be entitled to recover all costs (including reasonable attorneys' fees) which it or they may incur in connection with defending its interests and enforcement of legal rights arising due to a breach of this agreement by Company.

6. Need to Know. Company shall restrict disclosure of such Confidential Information to its employees and/or consultants with a need to know (and advise such employees of the obligations assumed herein), shall use the Confidential Information only for the purposes set forth in the Agreement, and shall not disclose such Confidential Information to any affiliates, subsidiaries, associates and/or third party without prior written approval of the disclosing party.

7. Intellectual Property Rights Protection. No license to a party, under any trademark, patent, copyright, design right, mask work protection right, or any other intellectual property right is either granted or implied by the conveying of Confidential Information to such party.

8. No Conflict. The parties represent and warrant that the performance of its obligations hereunder do not and shall not conflict with any other agreement or obligation of the respective parties to which they are a party or by which the respective parties are bound.

9. Authority. The parties represent and warrant that they have all necessary authority and power to enter into this Agreement and perform their obligations hereunder.

10. Dispute Resolution. If any difference or dispute arises between the Department and the

Company in connection with the validity, interpretation, implementation or alleged breach of any provision of this Agreement, any such dispute shall be referred to the Hon. Municipal Commissioner, JMC.

a) The arbitration proceedings shall be conducted in accordance with the (Indian) Arbitration and Conciliation Act, 1996 and amendments thereof.

b) The place of arbitration shall be Jamnagar.

c) The arbitrator's award shall be substantiated in writing and binding on the parties.

d) The proceedings of arbitration shall be conducted in English language.

e) The arbitration proceedings shall be completed within a period of 180 days from the date of reference of the dispute to arbitration.

11. Governing Law. This Agreement shall be interpreted in accordance with and governed by the substantive and procedural laws of India and the parties hereby consent to the exclusive jurisdiction of Courts and/or Forums situated at Jamnagar, India only.

12. Entire Agreement. This Agreement constitutes the entire understanding and agreement of the parties, and supersedes all previous or contemporaneous agreement or communications, both oral and written, representations and understandings among the parties with respect to the subject matter hereof.

13. Amendments. No amendment, modification and/or discharge of this Agreement shall be valid or binding on the parties unless made in writing and signed on behalf of each of the parties by their respective duly authorized officers or representatives.

14. Binding Agreement. This Agreement shall be binding upon and inure to the benefit of the parties hereto and their respective successors and permitted assigns.

15. Severability. It is the intent of the parties that in case any one or more of the provisions contained in this Agreement shall be held to be invalid or unenforceable in any respect, such provision shall be modified to the extent necessary to render it, as modified, valid and enforceable under applicable laws, and such invalidity or unenforceability shall not affect the other provisions of this Agreement.

16. Waiver. If either party should waive any breach of any provision of this Agreement, it shall not thereby be deemed to have waived any preceding or succeeding breach of the same or any other provision hereof.

17. Survival. Both parties agree that all of their obligations undertaken herein with respect to Confidential Information received pursuant to this Agreement shall survive till perpetuity even after any expiration or termination of this Agreement.

18. Non-solicitation. During the term of this Agreement and thereafter for a further period of two (2) years Company shall not solicit or attempt to solicit Department's employees and/or consultants, for the purpose of hiring/contract or to proceed to conduct operations/business similar to Department with any employee and/or

consultant of the Department who has knowledge of the Confidential Information, without the prior written consent of Department.

This section will survive irrespective of the fact whether there exists a commercial relationship between Company and Department.

19. Term. Subject to aforesaid section 17, this Agreement shall remain valid up to years from the “effective date”.

IN WITNESS HEREOF, and intending to be legally bound, the parties have executed this Agreement to make it effective from the date and year first written above.

For Department

Name:

Title:

WITNESSES:

1.

2.

For Company

Name:

Title:

WITNESSES:

1.

2

12 Annexure VIII: Sample Change Control Note

Change Control Note Initiation	CCN Number
Details of proposed change:	
Title: Originator: Date of Initiation:	
(To include reason for change and appropriate details/specifications. Please append attachments if any)	
Authorized by JMC Name: Signature: Date:	Received by SI Name: Signature Date:
To include the following: 1. A description of the change; 2. A list of deliverables required for implementing the change; 3. A timetable for implementation; 4. An estimate of any proposed change; 5. Any relevant acceptance criteria; 6. An assessment of the value of the proposed change; 7. Material evidence to prove that the proposed change is not already covered within the scope of the project, SLAs.	
Request for Authority to proceed	
Approved	
Rejected	
Requires Further Information	
(as follows or as attachments)	
For JMC	For selected SI
Signature	Signature
Name	Name
Title	Title
Date	Date

13 Annexure IX: Pre-Bid Query Format

SI requiring specific points of clarification may communicate with Home Department during the specified period using the following format:

SI 'S REQUEST FOR CLARIFICATION	
<<Name of Organization submitting query / request for clarification>>	
<<Full formal address of the Organization including phone and Email points of contact>>	Tel:
	Email:

Sr. No.	Page No.	Section No.	Point No.	Existing Clause	Clarification/Query of SI
.					

Please prepare the table in Excel Format as shown above. Any other form shall not be entertained.

14 Annexure X: Irrevocable Undertaking regarding GST

(On INR 500/- stamp paper)

I Shri / Smt. aged, years
Indian Inhabitant.

Proprietor/ Partner / Director of
M/s..... resident at do hereby give
Irrevocable undertaking as under;

1. I say & undertake that as specified in section 171 of CGST Act, 2017, any reduction in rate of tax on supply of goods or services or the benefit of input tax credit shall be mandatorily passed on to JMC by way of commensurate reduction in prices.

2. I further say and undertake that I understand that in case the same is not passed on and is discovered at any later stage, JMC shall be at liberty to initiate legal action against me for its recovery including, but not limited to, an appeal to the Screening Committee of the GST Counsel.

3. I say that above said irrevocable undertaking is binding upon me / my partners / company / other Directors of the company and also upon my / our legal heirs, assignee, Executor, administrator etc.

4. If I fail to compliance with the provisions of the GST Act, I shall be liable for penalty / punishment or both as per the provisions of GST Act.

Whatever has been stated here in above is true & correct to my / our own knowledge & belief.

Solemnly affirmed at

DEPONENT

This day of

BEFORE ME

Interpreted Explained and Identified by me

15 Annexure X: COTS GIS Platform Specifications

Specifications for COTS GIS Enterprise Platform

1. The proposed single deployed GIS Enterprise Platform should be Industry standard COTS GIS platform(16 Core). The proposed single deployed GIS platform should have all functions of GIS and Image analysis, geo-processing, classification, change detection, in a installed OEM platform only. The platforms should also have advance modules like network analysis, terrain analysis, 3D analysis, change Detection, Raster GIS analysis, SAR processing (Indian Based Sensors i.e. RISAT 1 &EOS-04 - Min Requirement) and 3D analysis.
2. The Proposed solution should be an integrated solution only having GIS, 3D Analytic and Image Processing components in single COTS Enterprise software. The overall proposed solution should be of Single GIS Platform only
3. The GIS Enterprise platform should have both GIS and Image analysis components, which going to be deployed at site. It should support 64 bits. It should based on a Services Oriented Architecture (SOA). The Enterprise GIS software must be OGC certified and must have the capability to serve and consume OGC complied web services including WMS, WFS, WCS, CSW, INSPIRE, etc. Licensing policy should not be restricted by hardware/ period or sites. The licensing policy should be open and do not need any additional cost / authorization process for redeploying it in new machine / cloud / VM. This shall provide flexibility to end user. No restriction during hardware upgrade/ scaling in future. OEM undertaking should be submitted for further customization of application based on GIS, 3D and Image processing Custom based web application, as per project needs of present and future. No restriction during hardware upgrade/ scaling in future.
4. Server deployed Software should run as a native 64-bit application and should support Windows / Linux 64- bit.
5. The deployed enterprise platform should have capabilities like geo-processing and analysis, spatial and statistics analysis and Image analysis functions. It should support server end Geo processing and Image analysis functions i.e. supervised classification, Enhancement, Vegetation Indices,Normalized Burn Ratio Index (NBRI),Normalized Difference Built-up Index (NDBI), buffer, clip, erase.

Provision should be available by server for sending the request using the web client and display the processed data on web using OGC certified services. No dependency of Desktop tools or workflow generated from desktop.

6. The web GIS application should open on any browser and should support cross platform. It should have RestAPI for integration with other applications
7. The GIS Enterprise platform should provide a map centric for managing geospatial content of organization. It should serve as project centric approach to Create, Access, Analyse, Manage and disseminate geo-spatial content with capability of user management and role-based access control.
8. The 3D components in Enterprise platform should have ability for 3D representation of the city or town with LOD1 to LOD4 level details as required. The 3D platform should have functionality for 3D analysis tools such as Measurement (Distance, Area, etc.), Dynamic based View shed, Line Profile, Line of Sight, Dynamic Contour, Flooding simulation, Dynamic Buffer generation of all geometry types. Capability to create slope & Aspect from terrain model, dynamic water, Dynamic River course, sub-merge analysis, Volumetric analysis. The software should have capability to create skyline based on observer position and distance. Model Cut tool for the visualization of object from different directions,
9. The GIS enterprise software should support draping of vector and raster: WMS, WMTS, KML/KMZ. Facilities to view Shape File, DGN, GPX & Geo-tiff, TIF. Provision to process and display CEOS SLC and Hybrid Polarimetric data of SAR, DTED, DEM. It should support various data such as Terrain (DEM, Tiff, DTED etc..), BIM & CAD (Revit, CAD, 3D-Max, Sketch-up, Maya etc.), 3D Building and animation file i.e CZML Collada - *.dae, *.obj, *.gltf. Photogrammetry Meshes data derived from UAV. Display of model generated from Drone, Satellite Stereo Pairs, LiDAR, Point Clouds (LiDAR data *.las, *.laz). Support of Vector data (GeoJSON, KML, Shape etc.), OGC (WMS, WFS, WCS, etc.), Capable of consume multiple third party OGC map services for visualization
10. It should have capability to import / export data in various formats like .dwg,dxf, .dgn, .shp (shape files), coverage file, .mif (MapInfo), .gml, .kml, .gpx, , Geo PDF GeoJSON, H4, H5 formats, MBtiles etc. Should support ODBC compliance

interface with industry standard RDBMS like PostGRE SQL, Oracle, SQL server, Access etc.

11. The deployed GIS enterprise Platform should have capabilities like geo-processing and analysis, spatial and statistics analysis and Image processing functions in a single window. It should have out of box Geo processing and Image analysis functions like Clip, Erase, Spatial Join, Relate, Buffer,clip, erase, intersection, dissolve, union, summarize, Image classification for ex -supervised, High pass, Low Pass filter, NBRI,NDBI, 3D viewer with 3D based analytic tools as out of box functions.
12. Software should have the facility for reports generation, customized map layout, high resolution printing in different formats (jpg, pdf) with desired map scale and customized templates.
13. Should have capability to store the applied legend, colour, symbology and further applied in other GIS projects. Further, provision should be there to take backup of the project, export and import the custom annotation and labelling pattern from one project to other. Thematic mapping like Bivarte, Multi-bivarte, chart mapping should be available as out of box. Geocoding and Reverse Geocoding should be available.
14. Integrated deployed Platform should must support Time aware data for Trends / Time Series Analysis. Provision should be there for Online and Offline GPS integration.
15. It should network analysis module to perform Routing analysis, Service Area Analysis, etc. Capabilities like Dynamic Labelling and Rule based Labelling should be available
16. The software should have function to perform different operations like: mathematical, logical, string operations on the field of table of vector layer or selected objects of vector.
17. The integrated software should allow visualization of data in 2D/3D in enterprise application. Provision should be available to host the 3d project from to browser mode. Facilities to import Satellite images, LIDAR, Building model. The system should have facilities to store the 3d model which can be used in future without doing any pre-processing.
18. The GIS Enterprise platform should have out of box provision to generate / create feature on 3d enviourment of all geometry types. Facilities for creation on 2d and

3d symbol should be there. Additionally, provision to share the 3d scenes is also required.

19. It should have module for terrain analysis and 3D analysis to create/ view - slope/aspect, hill shade, elevation profile, topographic normalize, line of sight and view shed analysis.
20. It should have algorithm for surface generation such as heat maps, Linear, IDW and Krigging.
21. The proposed software should support HRSI (High Resolution Satellite Imagery) and low resolution satellite images (panchromatic & multispectral) such as IKONOS, Quick bird, Geoeye, Worldview, CARTOSAT, EROS, LISS-IV, LISS-III, AWIFS, RISAT-1 &EOS-04, KALPANA-1, INSAT3A, INSAT3D, PROVA-V,CEOS,ECW,JP2000, Sentinel, Radarsat, etc..
22. It should have enhancement algorithm such as Linear, Logarithmic, Histogram Equalize, Histogram Matching, Density Slice, Gaussian, Squire root, Tone Balancing etc.

Specifications for COTS GIS Desktop Platform

1. The proposed deployed COTS GIS Desktop Platform should be Industry standard COTS GIS platform. The proposed deployed GIS platform should have all functions of GIS and Image feature extraction, geo-processing, image mosaicking, sub setting, classification (supervised, unsupervised) change detection (In-built Image processing capability) in a software. The platform should also have advance modules like network analysis, terrain analysis, 3D analysis, Raster GIS analysis & SAR processing (Includes RISAT 1 &EOS-04 processing ,Hybrid polarimetric)
2. The deployed Platform should have capabilities like geo-processing and analysis, spatial and statistics analysis and Image processing functions in a single window. It should have out of box Geo processing and Image analysis functions like Clip, Erase, Spatial Join, Relate, Buffer, clip, erase, intersection, dissolve, union, polyline to polygon, summarize, polygon auto numbering based on line feature and direction, Image classification(supervised, unsupervised), Image segmentation , Recode, Change detection module, Photogrammetric module, Resolution Merge,

Radiometric correction , Enhancement , Layer stacking, Geo-referencing ,Collage Image, Mosaicking, High pass, Low Pass filter, Fast Fourier transformation & 3D viewer with 3D based analytic tools as out of box functions

3. The proposed software should support HRSI (High Resolution Satellite Imagery) and low resolution satellite images (panchromatic & multispectral) such as IKONOS, Quick bird, Geoeye, Worldview, CARTOSAT, EROS, LISS-IV, LISS-III, AWIFS, RISAT-1 & EOS-04, KALPANA-1, INSAT3A, INSAT3D, PROVA-V, etc..
4. The desktop platform should be capable to process the temporal or time series image data. To identify encroachment and monitor urban sprawl, software should provide change detection capabilities as out of box tools such as Basic Change Detection, Advance Change Detection, Auto Change Detection and Site Monitoring. The advance change detection capability should allow to ingest multiple input images to find the change. It should also handle the multi resolution satellite image along with mis-registration
5. Should have photogrammetric extension which should have facilities for automatic stereo model by using stereo data from sensors like cartosat , worldview , Astrium , Geoeye by using RPC /RPB information. Also, provision should be there for auto mosaicking of stereo tiles by using sensor information. Provision should be there to create DSM, DTM & Ortho rectify Images from stereo pair. Also provision should there to generate the accuracy report of stereo model.
6. The Desktop platform should have support for SAR processing and analysis. SAR sensors like Radarsat, Sentinel, and RISAT 1 & EOS-04(Indigenous Sensors) support is required. Processing components like Importing, Conversion, Calibration, Generating Phase Image, Stokes vector file generation, hybrid polarimetric map, burst mosaic for EOS-04 is required for RISAT and EOS-4 sensors. Analysis functions like change detection, wavelet fusion, Radar vegetation analysis, CFAR functions are required in SAR. The platform should have capability to process microwave data such as RISAT, Sentinel etc. and provision to create calibrated dB image for threshold based flood mapping..

16 Fraud and Corrupt Practices

The SI and their respective officers, employees, agents and advisers shall observe the highest standard of ethics during the Bidding Process. Notwithstanding anything to the contrary contained herein, the Authority may reject an Application without being liable in any manner whatsoever to the SI if it determines that the SI has, directly or indirectly or through an agent, engaged in corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice in the Bidding Process.

Without prejudice to the rights of the Authority under relevant Clause hereinabove, if an SI is found by the Authority to have directly or indirectly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice during the Bidding Process, such SI shall not be eligible to participate in any tender or RFQ issued by the Authority during a period of 2 (two) years from the date such SI is found by the Authority to have directly or indirectly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice, as the case may be.

For the purposes of this clause, following terms shall have the meaning hereinafter respectively assigned to them:

A. “corrupt practice” means

the offering, giving, receiving, or soliciting, directly or indirectly, of anything of value to influence the actions of any person connected with the Bidding Process (for avoidance of doubt, offering of employment to, or employing, or engaging in any manner whatsoever, directly or indirectly, any official of the Authority who is or has been associated in any manner, directly or indirectly, with the Bidding Process or the LOA or has dealt with matters concerning the Concession Agreement or arising there from, before or after the execution thereof, at any time prior to the expiry of one year from the date such official resigns or retires from or otherwise ceases to be in the service of the Authority, shall be deemed to constitute influencing the actions of a person connected with the Bidding Process) or save and except as permitted under the relevant sub clause, engaging in any manner whatsoever, whether during the Bidding Process or after the issue of the LOA or after the execution of the Concession Agreement, as the case may be, any person in respect of any matter relating to the

Project or the LOA or the Concession Agreement, who at any time has been or is a legal, financial or technical adviser of the Authority in relation to any matter concerning the Project,

B. “fraudulent practice” means a misrepresentation or omission of facts or suppression of facts or disclosure of incomplete facts, in order to influence the Bidding Process,

C. “coercive practice” means impairing or harming or threatening to impair or harm, directly or indirectly, any person or property to influence any person participation or action in the Bidding Process,

D. “undesirable practice” means (i) establishing contact with any person connected with or employed or engaged by the Authority with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Bidding Process or (ii) having a Conflict of Interest; and

E. “Restrictive practice” means forming a cartel or arriving at any understanding or arrangement among SI with the objective of restricting or manipulating a full and fair competition in the Bidding Process.

F. If JMC determines that the SI has engaged in corrupt, fraudulent, collusive, coercive or obstructive practices, in competing for or in executing the Contract, then JMC may, after giving 14 days’ notice to the SI, terminate the SI employment under the Contract and expel him from the Site, and the provisions of relevant Clause shall apply as if such expulsion had been made.

G. Should any employee of the SI be determined to have engaged in corrupt, fraudulent, collusive, coercive, or obstructive practice during the execution of the Works, then that employee shall be removed in accordance with relevant Clause.

For the purposes of this Sub-Clause:

i. “corrupt practice” is the offering, giving, receiving to soliciting, directly or indirectly, of anything of value to influence improperly the actions of another party,

ii. “Another party” refers to a public official acting in relation to the procurement process or contract execution. In this context, “public official” includes Financer staff and employees of other organizations taking or reviewing procurement decisions,

- iii. “fraudulent practice” is any act or omission, including a misrepresentation, that knowingly or recklessly misleads, or attempts to mislead, a party to obtain a financial or other benefit or to avoid an obligation,
- iv. “collusive practice” is an arrangement between two or more parties designed to achieve an improper purpose, including to influence improperly the actions of another party.
- v. “coercive practice” is impairing or harming, or threatening to impair or harm.

17 Disclaimer

The information contained in this e-tender document or provided to SI(s), whether verbally or in documentary or any other form, by or on behalf of the Jamnagar Municipal Corporation (JMC), hereafter also referred as “The JMC Authority “, or any of its employees or advisors, is provided to SI(s) on the terms and conditions set out in this e-tender and such other terms and conditions subject to which such information is provided.

This e-tender includes statements, which reflect various assumptions and assessments arrived at by the Jamnagar Municipal Corporation (JMC) in relation to the Project. Such assumptions, assessments and statements do not purport to contain all the information that each SI may require.

This e-tender may not be appropriate for all persons, and it is not possible for the Jamnagar Municipal Corporation (JMC), its employees or advisors to consider the investment objectives, financial situation and particular needs of each party who reads or uses this e-tender. The assumptions, assessments, statements and information contained in this e-tender may not be complete, accurate, adequate or correct. Each SI should therefore, conduct its own investigations and analysis and should check the accuracy, adequacy, correctness, reliability and completeness of the assumptions,

Assessments, statements and information contained in this e-tender and obtain independent advice from appropriate sources.

Information provided in this e-tender to the SI(s) is on a wide range of matters, some of which may depend upon interpretation of law. The information given is not intended to be an exhaustive account of statutory requirements and should not be regarded as a complete or authoritative statement of law.

The Jamnagar Municipal Corporation (JMC) accepts no responsibility for the accuracy or otherwise for any interpretation or opinion on law expressed here.

The Jamnagar Municipal Corporation(JMC), its employees and advisors make no representation or warranty and shall have no liability to any person, including any SI or SI, under any law, statute, rules or regulations or tort, principles of restitution or unjust enrichment or otherwise for any loss, damages, cost or expense which may arise from or be incurred or suffered on account of anything contained in this e-tender or otherwise, including the accuracy, adequacy, correctness, completeness or reliability of the e-tender and any assessment, assumption, statement or information contained therein or deemed to form part of this e-tender or arising in any way with Pre-qualification of SIs for participation in the Bidding Process. The Jamnagar Municipal Corporation (JMC) also accepts no liability of any nature whether resulting from negligence or otherwise howsoever caused arising from reliance of any SI upon the statements contained in this e-tender.

The Jamnagar Municipal Corporation (JMC) may, in its absolute discretion but without being under any obligation to do so, update, amend or supplement the information, assessment or assumptions contained in this e-tender.

The issue of this e-tender does not imply that the Jamnagar Municipal Corporation (JMC) is bound to select and short-list pre-qualified Applications for Bid Stage or to appoint the selected SI or Concessionaire, as the case may be, for the Project and the Jamnagar Municipal Corporation (JMC) reserves the right to reject all or any of the Applications or Bids without assigning any reasons whatsoever.

The Applicant shall bear all its costs associated with or relating to the preparation and submission of its Application including but not limited to preparation, copying, postage, delivery fees, expenses associated with any demonstrations or presentations which may be required by The Jamnagar Municipal Corporation (JMC) or any other costs incurred in connection with or relating to its Application. All such costs and expenses will remain with the Applicant and the Jamnagar Municipal Corporation(JMC) shall not be liable in any manner whatsoever for the same or for any other costs or other expenses incurred by an Applicant in preparation or submission of the Application, regardless of the conduct or outcome of the Bidding Process.